Mining Electronic Health Records (EHRs): A Survey

PRANJUL YADAV, MICHAEL STEINBACH, VIPIN KUMAR, and GYORGY SIMON,
University of Minnesota - Twin Cities

The continuously increasing cost of the US healthcare system has received significant attention. Central to the ideas aimed at curbing this trend is the use of technology in the form of the mandate to implement electronic health records (EHRs). EHRs consist of patient information such as demographics, medications, laboratory test results, diagnosis codes, and procedures. Mining EHRs could lead to improvement in patient health management as EHRs contain detailed information related to disease prognosis for large patient populations. In this article, we provide a structured and comprehensive overview of data mining techniques for modeling EHRs. We first provide a detailed understanding of the major application areas to which EHR mining has been applied and then discuss the nature of EHR data and its accompanying challenges. Next, we describe major approaches used for EHR mining, the metrics associated with EHRs, and the various study designs. With this foundation, we then provide a systematic and methodological organization of existing data mining techniques used to model EHRs and discuss ideas for future research.

CCS Concepts: • **Information systems** → **Decision support systems**; • **Computing methodologies** → **Artificial intelligence**; **Machine learning**; • **Applied computing** → **Health informatics**;

Additional Key Words and Phrases: Healthcare analytics, EHRs, data mining, machine learning, healthcare informatics

ACM Reference format:
Pranjul Yadav, Michael Steinbach, Vipin Kumar, and Gyorgy Simon. 2018. Mining Electronic Health Records (EHRs): A Survey. *ACM Comput. Surv.* 50, 6, Article 85 (January 2018), 40 pages.
https://doi.org/10.1145/3127881

1 INTRODUCTION

Numerous recent studies have found that the healthcare system in the United States is the most expensive in the world, yet it trails behind most advanced economies in quality (Schuster et al. 1998). The cost of healthcare is steadily rising, both in absolute terms and as a percentage of the GDP, and will soon reach unsustainable levels (Levit et al. 2003). To curb this trend, the US government has put several initiatives in place, aiming to simultaneously improve care quality and decrease its cost. Many believe that advanced analytics holds the key to achieving these opposing goals.

The earliest of these initiatives was the mandate for healthcare providers to implement electronic health record (EHR) systems. Although the primary goal of the EHR is to document patients' care for reimbursement, a frequent *secondary use* of the accumulated data is to serve as a research platform. This EHR data, coupled with other health-related data, provides the platform on which advanced analytics can be built.

The work described in this manuscript was supported by NIH grant LM011972 and NSF grants IIS-1344135 and IIS-1602394. The views expressed in this manuscript are those of the authors and do not necessarily reflect the views of NIH and NSF.
Authors' addresses: P. Yadav, M. Steinbach, V. Kumar, and G. Simon, 200 Union St SE, Minneapolis, MN 55455, USA.
Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
© 2018 ACM 0360-0300/2018/01-ART85 $15.00
https://doi.org/10.1145/3127881

The motivation for advanced analytics comes from another government mandate that transforms the current fee-for-service payment model to a new model based on population health management. Under the new model, primary care providers are responsible for managing entire patient populations, with their payments tied to care quality. Since care providers are no longer paid for services rendered, but rather for the outcomes of the service, they are incentivized to increase their efficiency through implementing better care practices, thus creating an opportunity for analytics.

With ample EHR data and a motivation for analytics, what is needed is an outlet for the findings. The learning healthcare system, a concept embraced by the Institute of Medicine, is an adaptive system, with "best practices seamlessly embedded in the delivery process and new knowledge captured as an integral by-product of the delivery experience." In this context, advanced analytics, clinical data mining included, will play a key role in extracting new knowledge, quantifying the effects of changes in care delivery, and potentially guiding the formation of best-practice guidelines.

The learning healthcare system initiative is accompanied by other symbiotic initiatives that are likely to influence the medicine of the future. Precision medicine is one of them. It is a model where treatment is tailored to each individual patient. While precision medicine has a large genomic component, finding increasingly specific treatments from EHR data is another avenue, one that naturally fits into today's practice framework.

Today's predominant medical practice framework is evidence-based medicine (Sackett 2000). At the heart of evidence-based practice lies a large knowledge base of best-practice recommendations that have been put forth by committees of well-established care providers and (ideally) validated through randomized clinical trials. These best-practice recommendations, if followed, can enable providers to increase their effectiveness while reducing wasteful practices.

The traditional vehicle for evidence creation is the randomized clinical trial (RCT) (Matthews 2006). Randomized clinical trials are considered the gold standard for evidence creation, but the body of evidence they create is woefully incomplete. This can be corroborated by the fact that RCTs are usually performed using small populations, and hence any meaningful results obtained have uncertain validity for the genetically and ethnically diverse global population. Trials are very expensive to conduct, and thus they are relegated to a *confirmatory* role. In particular, they are impractical for exploration.

Clinical data mining using EHR data offers a solution that is complementary to clinical trials. With EHRs tracking entire populations over extended time periods, EHR data has large sample sizes and potentially long follow-up times, which potentially allows testing of a large number of exploratory hypotheses. Using clinical data mining approaches, EHR data can provide useful information to facilitate clinical decision support, to analyze condition-specific clinical process outcomes, and to improve team-based population care beyond the traditional clinical encounters.

With all three ingredients of success for data mining—data, motivation, and outlets for results—having come together, we believe that the growth in the adoption of clinical data mining will accelerate. However, EHR data poses unique challenges that data mining has previously not faced. Before reviewing existing work on clinical data mining involving EHR data, we summarize background knowledge on study design, the characteristics and challenges of EHR data, and techniques developed by other disciplines to overcome these challenges. We hope that our survey will prove to be a useful tool for turning the challenges that EHR data poses into a wave of innovation in data mining that will unlock the immense potential of clinical data mining.

Overview: The structure of our survey reflects the steps carried out in clinical research. Our first step is to frame a clinical problem that we wish to study. In Section 2, we give the reader a flavor

of clinical questions by providing descriptions of several applications starting from the simplest clinical questions and progressing toward more complex analytics tasks.

Once the clinical question has been defined, data modeling starts. In Section 3, we provide a detailed description of EHR data and its elements and characteristics. As we discussed earlier, the primary function of EHR data was not to serve as a research platform, and thus it poses numerous challenges, which we describe in Section 4.

The next step in the analysis process selects a metric of interest. EHR data consists of multi-dimensional, multimodal irregular time-series data with many potential outcomes. A rich array of metrics can be designed and measured on such data, but only a handful are well defined, understood, and accepted in the health sciences research community. In Section 5, we describe the most commonly used metrics, that is, those that the community tracks and uses to inform policy decisions.

EHR data is a collection of data tables, but most data mining algorithms cannot directly operate on such a format. These tables need to be summarized into a data matrix in a manner that allows us to answer our clinical question or compute our metric of interest. This process is known as the creation of a study design, and in Section 6, we will introduce the study designs that are most suitable for mining EHR data.

At this point, we have a data matrix on which we could run our data mining algorithms, but doing so has the potential to produce misleading results. The EHR data still has some challenges that could seriously affect or invalidate our analysis. Examples of such challenges include situations when a patient drops out before the study concludes (referred to as censoring), patient populations whose members are in widely different states of health and whose outcomes are thus not comparable, and a highly variable number of per-patient observations. In Section 7, we introduce higher-level frameworks that have been developed to overcome such challenges.

Data mining algorithms can operate within these frameworks or can be adapted to fit into these frameworks. In Section 8, we provide a comprehensive overview of how data mining methods have been applied to mine EHR data and indicate how these data mining methods have contributed to the clinical research carried out in the various application areas that we introduced in Section 2.

In Section 9, we present the current state of clinical data mining and identify areas that are well covered by existing techniques and areas that require the development of new techniques. Finally, Section 10 concludes the survey with a discussion of the future of EHR data mining.

Although healthcare data mining is still in its infancy, this survey has covered several hundred scientific articles in this area. We describe the required background knowledge and lay out the principles that allow us to organize this vast body of research. Due to page limitations, we have focused the main article on the sequence of steps researchers need to take to complete an EHR-based data mining analysis citing only a small number of representative examples. Accompanying supplemental material contains a more comprehensive listing of papers describing applications of data mining to EHR data.

Scope: The primary purpose of this survey is to provide the necessary background and an overview of the existing literature of mining structured EHR data toward answering clinical questions. We decided to exclude areas such as bioinformatics, computational biology, and translational bioinformatics in order to narrow the scope of this survey. For the same reason, we also excluded visualization and privacy-preserving data mining. These areas are important but would draw focus away from the main theme of the survey. We also excluded medical image analysis and techniques related to mining semistructured or unstructured data through Natural Language Processing (NLP) and information retrieval because the technical challenges they pose are quite different from the challenges we face in mining structured EHR data.

2 APPLICATION AREAS

An appropriate place to start our discussion of clinical data mining is to describe some clinical questions that data mining can help answer. The fundamental question of medicine is to decide on the treatment that is most suitable and effective for a particular patient, and data mining has the potential to help address this question in numerous ways.

In this section, we present a sequence of increasingly complex clinical applications, starting with the simplest epidemiological questions; continuing with common clinical research tasks such as automatic identification of study cohorts, risk prediction, and risk factor discovery; and concluding with complex applications such as discovering knowledge related to best clinical practices from data.

2.1 Understanding the Natural History of Disease

The most basic epidemiological inquiries are not data mining problems per se, but are concerned with questions like: How many patients are diagnosed with a condition of interest? What is the incidence rate and symptoms of a disease? What is the *natural history of progression of the disease*, or in other words, what are the *symptoms and implications* of any clinical condition of interest?

Studying the sequences of diseases clearly enters the realm of data mining and it has been applied to studying the progression of a patient's medical state over time, which is also known as the patient's *medical trajectory*. Examples of such trajectories are the progression of the patient from a healthy state through conditions like hyperlipidemia (high cholesterol), hypertension (high blood pressure), and diabetes toward diabetes-associated complications (e.g., amputation, severe paralysis, or death). Often, multiple trajectories lead to the same outcome. For example, consider an outcome such as mortality. In this case, a patient might die due to kidney complications, cardiovascular complications, or peripheral complications. Even though the outcome (mortality in this case) is the same, disease progression paths leading to the outcome will typically be different. Research studies have demonstrated that different trajectories can have significantly different associated risks for the same outcome (Oh et al 2016; Yadav et al. 2015b). Studying such varying trajectories can lead to the development of tailored treatments, the discovery of biomarkers, or the development of novel risk estimation indices.

The number of patients suffering from two or more chronic diseases, a condition referred to as *multimorbidity*, has increased substantially in the last 10 years, and multimorbidity is the norm with elderly patients (Mercer et al 2009). Diseases co-occur due to coincidence, causal relationships, and common underlying risk factors, including frailty. The coexistence of multiple chronic diseases increases the risk of mortality, leads to loss of physical function, decreases quality of life, and is also associated with longer hospital stays, increased risk of postoperative complications, and a higher overall healthcare utilization (Wolff 2002). When diseases co-occur, it is desirable to treat them simultaneously. *Comorbidity analysis* is the process of exploring and analyzing relationships among diseases and is the key to understanding multimorbidity and tailoring treatment accordingly. An example of multimorbidity is type 2 diabetes mellitus (T2DM), which is often accompanied by hypertension, hyperlipidemia, and impaired fasting glucose (IFG); these latter three conditions are *comorbid* to T2DM. Further, analyzing the comorbidities and discovering the relationships among them can lead to the modification of existing comorbidity scores (such as the Charlson index) or to the development of novel ones.

2.2 Cohort Identification

Once we understand these fundamentals, we can try answering more advanced questions. To this end, we may need to assemble a cohort (group) of patients, some of whom are extremely likely

to have the disease of interest (*cases*) and others who most likely do not (*controls*). This can be achieved through phenotyping algorithms, either hand-crafted or machine learned. Phenotyping algorithms characterize the disease in terms of patient characteristics observable from the EHR data and classify patients as likely having the disease, likely not having the disease, or disease status is uncertain (Kirby et al. 2016).

Traditionally, cohort identification was carried out through chart reviews, where nurse abstractors painstakingly reviewed patients' medical records to identify whether each patient satisfied the criteria for inclusion into the cohort. However, the scale enabled by EHRs renders manual chart review impractical. Instead, electronic phenotyping algorithms are applied, with manual chart review relegated to spot-checking. Cohort identification has been widely used in various clinical research studies and biomedical applications, forming the platform for future studies in areas such as pharmacovigilance, predicting complications, and quantifying the effect of interventions.

A phenotype is defined as a biochemical or physical trait of an organism, such as a disease, physical characteristic, or blood type. Examples of phenotypes in EHRs are clinical conditions, characteristics, or sets of clinical features that can be determined solely from the EHR data. Such techniques are useful for identifying patients or populations with a given clinical characteristic from EHRs using data that are routinely collected and stored in disease registries or claims data. Phenotyping queries used for cohort identification can be used at different sites in a similar fashion in order to ensure that populations identified across healthcare organizations have a similar clinical state. Phenotypic definitions can also be used for direct identification of cohorts based on risk factors, clinical or medical characteristics, and complications, thereby allowing clinicians to improve the overall healthcare of a patient. For a detailed review of phenotyping, the reader is referred to Shivade et al. (2014).

2.3 Risk Prediction/Biomarker Discovery

With a cohort in hand, we can build risk models, opening up a wide range of opportunities for data mining. An example of a successful risk model is the Framingham heart score, which estimates patients' risk of cardiovascular mortality. In recent years, age-adjusted cardiovascular deaths have reduced by half in developed countries. Much of this success is attributed to the Framingham Heart Study, which helped identify the key risk factors of cardiovascular mortality (Bitton and Gaziano 2010). This success can potentially be replicated in other areas of medicine.

Such models can predict the risk of developing a disease, for example, estimating the probability of developing a condition of interest in 5 years (*risk prediction*). Such analysis is often performed to identify high-risk individuals and facilitate the design of their treatment plans (Ng et al. 2014). Interventions prescribed by risk analyses can lead to improvement in a patient's health, thereby preventing the patient from progressing to advanced complications.

In some cases, predicting the patient's risk of progression is secondary to understanding the underlying risk factors. Risk models can provide information about the importance of risk factors. Risk prediction also provides the opportunity to identify significant indicators of a biological state or condition. In simple terms, a biomarker is defined as a set of measurable quantities that can serve as an indicator of a patient's health. Biomarkers offer a succinct summary of the patient's state with respect to a medical condition. Rather than having to analyze the thousands of variables present in an EHR, it can be sufficient to focus on relatively few biomarkers to paint a reasonably accurate picture of the patient's overall health. Over the years, biomarkers have found numerous applications. For example, abnormal hemoglobin A1C (a measure of blood sugar) is a biomarker for type 2 diabetes mellitus (T2DM) and high cholesterol is a biomarker for being at risk of cardiovascular complications. Similarly, there are certain biomarkers that are common across many diseases. For example, age is by far the most common biomarker. It indicates that as a person ages,

his or her risk to acquire certain diseases (e.g., T2DM, cardiovascular complications, and kidney complications) increases significantly. EHRs provide a platform to identify, analyze, and explore biomarkers for multiple diseases.

2.4 Quantifying the Effect of Intervention

Interventions are often drug therapies, procedures, or surgeries, but can also include recommendations for lifestyle changes and/or patient education. Choosing the optimal treatment for a patient requires us to be able to estimate the effect of the possible interventions. Specialized data mining methods such as uplift modeling or statistical techniques in combination with causal analysis can be used to *quantify the effects of interventions*.

The longitudinal aspect of EHRs provides an opportunity to analyze the effects of intervention for longer periods of time across larger cohorts. It also provides clinicians with a platform to analyze whether the interventions have any accompanying adverse effects. Moreover, EHRs provide a platform to analyze whether interventions vary across cohorts based on demographic attributes such as gender, age, ethnic makeup, socioeconomic status, and so forth.

2.5 Constructing Evidence-Based Guidelines

Once the effect of a treatment has been proven, this knowledge can be codified into and disseminated as clinical practice guidelines. *Evidence-based clinical practice guidelines* are considered the cornerstone of modern medicine, and they provide guidance on the optimal treatment under a particular set of conditions based on epidemiological evidence.

Clinical guidelines can be defined as standardized specifications and procedures usually used to improve or take care of a patient in specific clinical circumstances (Field et al. 1992). Evidence-based guidelines try to guide decision making by identifying best clinical practices that are meant to improve the quality of patient care (Barretto et al. 2003). They help clinicians make sound decisions by presenting up-to-date information about best practices for treating patients in a particular medical state, including expected outcomes and recommended follow-up interval. For example, the American Diabetes Association (ADA) guidelines for diabetes consist of recommendations for diagnosing the condition (e.g., a patient is considered diabetic if his or her hemoglobin A1c is >6.5), controlling the disease (patient is under control if his or her A1c is <6.5 and systolic blood pressure is <140mmHg), and prescribing interventions (lifestyle modification and therapeutic interventions). These guidelines are often regarded as the cornerstone of modern healthcare management.

2.6 Adverse Event Detection

This term describes the detrimental effect of medical care on patient medical state. Examples include infection acquired during the treatment of a different condition, such as surgical site infection. According to a 2010 report by the inspector general of the Department of Health and Human Services, 13.5% of hospitalized Medicare beneficiaries have experienced adverse events, costing Medicare an estimated $340 million in October 2008 alone. An estimated 1.5% of beneficiaries, which corresponds to 15,000 patients per month, experienced an adverse event that led to their deaths (Levinson and General 2010). Detecting and learning to prevent adverse events is a major opportunity for advanced analytics, as 44% of these adverse events were deemed avoidable.

Another related opportunity is the identification of potentially preventable events. Potentially preventable events are patient encounters (emergency and urgent care visits or hospitalization) that could have been avoided by appropriate outpatient treatment and adequate compliance with those treatments. The Agency for Healthcare Research and Quality (AHRQ) reports that potentially

preventable events are decreasing, but still nearly 4 million hospitalizations in the United States were avoidable in 2010 if the patient had received proper care (Torio et al. 2006). Costs related to these preventable hospitalizations totaled $31.9 billion in 2010.

Adverse events can often be linked to drugs. Adverse drug events account for one in three hospital adverse events, and in the outpatient setting, adverse drug events cause 3.5 million physician visits, 1 million emergency room visits, and 125,000 hospitalizations each year (Torio et al. 2006). Although drugs are tested for any potential adverse effects before they are released for widespread use, often test cohorts are small with short observation periods. Several agencies conduct research on detecting adverse drug reactions: the US Food and Drug Administration with its adverse event reporting system, the European Medicines Agency, and the World Health Organization, which maintains an international adverse reaction database. Unfortunately, all these agencies suffer from underreporting and biased analyses of adverse drug reactions. EHRs provide a new platform to improve and complement drug safety surveillance strategies (Stang et al. 2010).

3 THE NATURE OF EHR DATA

One motivation behind the federal mandate for EHRs was to document patients' state of health over time and the therapeutic interventions to which these patients were subjected. EHRs store this information in structured (databases), semistructured (flow sheets), and unstructured formats (clinical notes). The format of the information greatly affects the ease of access and quality of the data, and thus has a substantial impact on the downstream data mining.

3.1 Structured Data

From the viewpoint of healthcare analytics, retrieving structured data is a straightforward task. Structured data is stored in database tables with a fixed schema designed by the EHR vendor. The most commonly used information, such as demographic information (e.g., birth date, race, ethnicity), encounters (e.g., admission and discharge data), diagnosis codes (historic and current), procedure codes, laboratory results, medications, allergies, social information (e.g., tobacco usage), and some vital signs (blood pressure, pulse, weight, height) are all stored in structured tables. This kind of information is common across providers and not specific to any clinical specialty. The use of structured data tables with a priori defined layouts (schema) enables high performance (rapid access to data) and standardization: the schemas for these tables are very similar if not identical across installations by the same EHR vendor, thus requiring very little (if any) site-specific knowledge from users. This quasi-standardization of fields also greatly helps information retrieval for analytic purposes.

Storing all information in EHRs as structured elements, however, is impractical: it would require anticipation of all possible data elements (e.g., metrics whose usefulness we do not yet appreciate) and would result in a level of complexity that would render the EHR system unusable. Hence, there is a need for storing information that does not readily fit into the admittedly rigid schema of the structured tables. For example, clinicians often write notes about patients' symptoms based on their previous experiences, which is hard to standardize a priori.

3.2 Unstructured Data

Among the three formats, clinical notes (unstructured data) offer maximal flexibility. Clinical notes mostly store narrative data (free text). Many types of clinical notes are in existence, and the information that resides in them depends on the type of note (e.g., radiology report, surgical note, discharge note). These clinical notes can contain information regarding a patient's medical history (diseases as well as interventions), familial history of diseases, environmental exposures, and

lifestyle data. NLP tools and techniques have been widely used to extract knowledge from EHR data.

Clinical notes such as admission, treatment, and discharge summaries store valuable medical information about the patient, but these clinical notes are very subjective to the doctor or the nurse writing them and lack a common structure or framework. These clinical notes also have grammatical errors, short phrases, abbreviations, local dialects, and misspelled words. Considerable data processing needs to be conducted on these clinical notes, such as spelling correction, word sense disambiguation, contextual feature detection, extraction of diagnosis codes from clinical text, and adverse events surveillance. This makes deriving information about patient characteristics from clinical notes a computationally challenging task that requires the most sophisticated NLP tools and techniques. For a detailed review on applications of NLP to clinical decision support, the reader is referred to Demner-Fushman et al. (2009).

3.3 Flowsheets

In between the two extremes (structured tables and unstructured clinical notes) lies the (semistructured) flow sheet data format. This format is most reminiscent of resource description files (RDFs), consisting of name, value, and timestamp triplets. Typically, the "name" field stores the name of the measure and the "value" field contains the actual measurements: for example, the name is "arterial blood pressure" and the value is 145mmHg. This format is more flexible than the structured tables, since the user can define new metrics through the name field; the set of metrics is not restricted to those anticipated by the EHR vendor. Flow sheets are similar to structured data in the sense that the value field is either a quantitative measure (e.g., blood pressure) or typically a restricted set of values. For instance, the American Society of Anesthesiologists (ASA) physical status takes values of "mild systemic disease," "healthy," "severe systemic disease," "moribund," or "severe life-threatening systemic disease."

Flow sheets offer expandability to EHR systems and thus have found numerous uses, becoming the only or most convenient data repository for many applications. Possibly the most important use for flow sheets is that they provide detailed information about specialty care. Examples include information related to a patient's asthma care plans and various diabetes-related nonstandard (or not-yet-standard) metrics for a diabetes clinic. In addition, they may provide additional details regarding how a particular measure was obtained (blood pressure taken while the patient was lying flat) and can also be used to store automated sensor data (e.g., pulse and blood oxygen levels every few minutes in an intensive care unit). Further, flow sheets can be used to pull together related measurements such as quality indicators.

4 DATA-RELATED CHALLENGES

EHR data as a research platform poses numerous challenges. Many of those challenges are not specific to EHR data and are also frequently encountered in other domains: noise, high dimensionality, sparseness, nonlinear relationships among data elements, and complicated dependencies between variables. Less frequent, but still not uncommon elsewhere, are issues related to data integration across multiple sites (medical providers) and/or multiple types of datasets (e.g., clinical vs. claim data). As in other domains, it is also important to incorporate domain knowledge, including knowledge about the relationships. For example, the blood pressure of a patient on medication for hypertension needs to be interpreted in that context.

In this section, we describe a number of challenges, common in healthcare but less studied in other areas, that put the external validity of the analysis at risk. Many of these relate to issues stemming from data missing for various reasons.

4.1 Censored Data

By censored data, we refer to the situation where a patient's state is only observable during a certain period of time, or conversely, when potentially interesting events fall outside the observation period and are hence unobservable. In case of left-censored data, patients experience events of interest prior to the start of the study; in case of right-censored data, potentially interesting events are unobservable because they happened to the patient after he or she dropped out of the study or after the conclusion of the study. In case of interval-censored data, the only information available is that the event of interest happened within a certain period of time. Studies can be either left, right, or interval censored. Censoring can lead to loss of crucial information about the patient's health. For example, for the right-censored patient, there is neither an easy way to determine whether the patient is alive or dead nor a way to measure the efficacy of the treatment the patient was undergoing.

4.2 Fragmentation

Fragmentation is a lack of data sharing across providers. Fragmentation typically occurs when patients visit multiple healthcare providers seeking specialty care, expert advice, or second opinions. In such scenarios, all healthcare providers involved only have partial information about the patient's medical history. Integrating data across multiple healthcare providers poses several challenges. Some of these are operational challenges, such as communication between different EHR systems such as General Electric (GE) or Epic. Such communication requires a common language to transfer information, such as HL-7, an international standard protocol designed by healthcare providers to share clinical data across software applications (Rodrigues 2009). Unfortunately, this approach cannot capture all nuances. Even when multiple sites use the same EHR format, their treatment policies may differ, flowsheets may differ, and thus their definitions of nuanced concepts may differ. For example, fasting and random glucose measurements are not distinguished by lab codes, and different sites can apply different methods to distinguish the two. Other challenges involve the (lack of) willingness of competing providers to share data.

4.3 Irregular Time-Series Data

Besides our inability to make observations before the study period starts or after it concludes, the most striking characteristic of the EHR data is the irregularity of the patient visits. While a recommended frequency of visits may exist, many patients do not actually follow these recommendations. For example, as per the American Diabetes Association (ADA) guidelines, the A1C test must be performed at least two times a year for patients who are meeting treatment goals and have stable glycemic control.

Further, the frequency at which information is collected varies. While vitals are collected at every visit, certain laboratory tests are ordered annually, and other tests are performed only as needed. This difference in the frequency of collection of medical information leads to irregular longitudinal data. Analyzing regular time series is a well-studied problem in data mining and other fields, but application of these techniques to EHR-type irregular time series is very challenging.

4.4 Other Sources of Missing Data

Diagnosis codes might also be missing due to reimbursement rules. Different problems, comorbidities, or complications have different reimbursement rates: depending on the complications, the same procedure may have different costs and thus result in increased or decreased reimbursements. Due to these financial constraints, only some of the problems related to the primary cause of the visit are used to generate billing codes. (ICD codes of various versions are currently used to

represent medical diagnoses in billing records.) This leads to biases, as the billing codes might not be a true representation of the actual medical state of the patient.

Diagnosis codes can also be missing due to changes in disease definitions and updates in the coding scheme, such as the recent change from ICD-9 to ICD-10 codes. For example, prediabetes did not have a corresponding ICD code until 2000. The introduction of new and periodic updates to existing ICD codes lead to further complications such as a lack of a clear mapping from the old revision to the new one and, subsequently, to inconsistent research findings.

Another largely unobservable source of missing information lies in patient conformance with prescriptions and other intervention, such as lifestyle change recommendations. The orders table in an EHR indicates that the physician prescribed a medication, but in most cases we do not know whether the patient actually took the medication. This situation is referred to as Intent to Treat. In the case of a lifestyle change, we may not even have documentation that the patient followed this advice.

A unique aspect of missing data in clinical analytics is that whether the data is missing or not can be predictive. When a physician orders a test, he or she usually suspects that the patient may suffer from the corresponding condition. Conversely, by not ordering certain tests, the clinician suggests that corresponding medical conditions are absent. For example, no bacterial panel being ordered likely indicates that the patient is not suffering from such an infection.

4.5 Biases and Confounding Effects

Studies performed using EHRs often have biases (Hripcsak et al. 2011) and confounding effects (Moher et al. 1998). Biases might arise for multiple reasons. For example, in a cohort study (Section 6.1), there might be significant differences in baseline characteristics (age, gender, race, ethnicity) between the cases and the controls (Gruber et al. 1986). In such cases, any observed difference between the groups after a follow-up period might be due to the difference in baseline characteristics and not due to the exposure, and thus, quantifying the real effect of exposure might be difficult.

Such bias can be overcome by finding the right control group. One possible way is to randomly select subjects from a pool of patients such that the pool does not consist of patients diagnosed with the outcome (Wacholder et al. 1992). In other approaches, controls can be drawn from the neighborhood of the cases, as such controls would be very similar in terms of socioeconomic status and lifestyle choices (Vernick et al. 1984). Similarly, when genetic factors are the main focus of study, controls could often be chosen from family and relatives, as they share a similar genetic makeup (Wacholder et al. 1992).

Confounding is another issue that might undermine the internal validity of any study (Abramson et al. 2001). Confounding arises when a variable (i.e., a confounder) is associated with both the exposure and outcome variables and thus creates a spurious relationship between the two (Ory 1977). For example, studies have often reported a high degree of association between risk of myocardial infarction and oral contraceptives. However, this association was later found to be spurious because of the high proportion of tobacco users among users of birth control pills. Therefore, tobacco consumption confounded the relation between myocardial infarction and oral contraceptives.

5 METRICS

Quantifying the outcome or the effect of an exposure on the outcome are the primary objectives in many research studies. The most common metrics are incidence rate, prevalence, relative risk, and odds ratio (Hennekens et al. 1987). Incidence rate (Last et al. 2001) indicates the number of new cases of a disease in a population at risk over a predefined interval of time, and prevalence indicates

the number of existing cases of the disease in a population under observation. For example, in 2012, 1.4 million Americans developed diabetes, making the total number of Americans with diabetes 29M. With the US population being 312M at that time, the prevalence of diabetes was $29/312 = .09$ (9%). Since 28M people of the 312M already developed diabetes before 2012, there were $312 - 28 = 284$M people at risk, resulting in an incidence rate of $1.4/284 = .005 = 0.5\%$ for 2012, or 5 in 1,000 person-years (5/1,000py).

Relative risk is defined as the frequency of the outcome in the exposed group as compared to the frequency of the outcome in the unexposed group. For example, consider a cohort of 4,563 prediabetic patients, in which 1,865 (41%) take statin (exposed group) and the remaining 2,698 patients do not (Castro et al. 2016). The cohort is then followed for 6 years. Within these 6 years, 682 patients in the exposed group progressed to diabetes, whereas 842 patients in the control group progressed to diabetes. From this information, the relative risk of diabetes is $(682/1865)/(842/2698) = 1.17$; that is, patients in the treatment group are 1.17 times as likely to progress to diabetes as the controls. Relative risk is 1.0 when the frequency of the outcome is the same in both the groups. Relative risk greater than 1.0 indicates an increased risk of outcome in the treatment group, while less than 1.0 indicates a decreased risk (protective effect of the exposure).

Odds ratio can be defined as the odds of exposure/outcome in the exposed group divided by the odds of the exposure/outcome in the control group. For the previous example, the odds ratio is $(682/1183)/(842/1856) = 1.27$, where 1,183 (and 1,856) are the number of patients in the exposed group (and the control group, respectively) who did not develop diabetes. In the following section, we describe various study designs and metrics that are widely used in those study designs.

6 STUDY DESIGN

EHR data is a mere collection of database tables that need to be transformed into an analysis matrix that is amenable to data mining in a manner that allows us to answer the question we set out to study. Suppose we wish to construct a risk prediction model that can predict the 5-year risk of a particular disease D for any patient. To construct such a model, we would take a cross-section of the patient population in year Y, allowing us to have a representative sample from which this "any patient" may come from. Y is ideally at least 5 years before the last date in the EHR, so that we have sufficient (5-year) follow-up for many patients. Next, we reconstruct all patients' state of health in year Y, collecting their medical history before Y, and then follow them 5 years forward (until year $Y + 5$) and establish their outcomes, namely, whether they developed D. The analysis matrix would have patients as its rows and patient characteristics in year Y as columns, and would have an outcome for D as well. Traditional predictive modeling techniques are directly applicable to such a matrix for carrying out various research related tasks.

The way we transformed the EHR data follows a particular *study design* that allows us to answer our question. Each study question can require a different study design. In this section, we review some of the most commonly used study designs and the questions they allow us to answer. For additional details, the interested reader is referred to Grimes and Schulz (2002).

Study designs form a hierarchy, which is depicted in Figure 1. Accordingly, at the highest level, study designs can be primarily classified into two major groups, that is, experimental and observational. In an **experimental** study design, the researcher intervenes to change the course of the disease and then observes the resultant outcome. Randomized clinical trials are examples of experimental study designs. A specific example would be a study where surgery patients with T2DM were randomized to receive supplemental insulin at bedtime for blood glucose (treatment) or no supplemental insulin (control). As intervention in EHRs is not possible, we will not discuss these study designs any further.

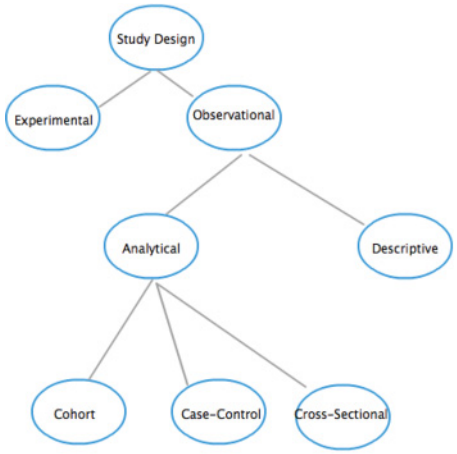


Fig. 1. Study design classification hierarchy.

By **observational** (Funai et al. 2001), we refer to study designs where the researchers do not intervene. In such studies, the treatment allotted to each subject is beyond the control of the investigator. For example, consider a study that investigates the effect of smoking (exposure) on lung capacity (outcome). A cohort of young men aged 18 to 25 are identified. Some subjects in this cohort smoke tobacco (exposed group) and some do not (unexposed/comparison group); the investigator has no influence on which subjects smoke and which do not. This cohort is then followed for a number of years to analyze the effect of smoking on lung capacity by comparing the exposed group with the unexposed group (Kelsey 1996; Rothman and Greenland 1986). Observational studies can be further categorized as analytical (if there is a comparison group as in the example given earlier) or descriptive (no comparison group).

Analytical studies are mostly used to test hypotheses about exposures and their effects on the outcome. They can also be used to identify risk and protective factors for diseases as well as causal associations between exposures and outcomes. Analytical studies (Last et al. 2001) can be further divided into three major groups based on the temporal direction in the study. Studies that start with an outcome and look back in time for exposure are known as **case-control studies**. If the study begins with an exposure and concludes with an outcome, we refer to them as **cohort studies**. If we only consider a single point in time, where the outcome and the exposure may both be present at the same time, we refer to the study as **cross-sectional** (Last et al. 2001).

Descriptive study designs, which have no outcome of interest, mostly deal with the frequency and the distribution of risk factors in populations and enable us to assess the extent of a disease of interest. These study designs are usually used to build hypotheses, thereby building the framework for future clinical research.

In Figure 1, we categorized studies as analytical or descriptive first and then categorized them further based on temporal direction. Alternatively, and equally correctly, we could have categorized them based on temporal direction first. Studies that start with a cohort selection and follow the cohort forward in time are called **prospective**; studies that start with the outcome and look backward in time are **retrospective**; and studies that take a snapshot of the patient population at a particular point in time are **cross-sectional**. Within each of these categories, we could further divide them as analytical or descriptive. For example, a case-control study is a retrospective analytical study. Retrospective descriptive studies are also possible.

In the following section, we will discuss the aforementioned study designs along with clinically relevant examples. We will also discuss how certain study designs may introduce biases and confounding factors.

6.1 Cohort Studies

Cohort studies are also known as incidence, follow-up, forward-looking, longitudinal, prospective, or concurrent studies (Lilienfeld and Stolley 1994). The defining feature of these studies is that we follow a cohort of patients over time. The cohort consists of two kinds of patients: those who are exposed to a particular factor of interest (called the *exposed* group) and those who are not exposed (*unexposed* or *control* group) at a particular point in time (*baseline*). We follow this cohort and compare the incidence of an outcome between the exposed group with the unexposed group. Often the exposure of interest is a treatment and researchers are interested in assessing the effect of a treatment on an outcome. In this case, the exposed group can also be called the *treatment* group.

As an illustration, let us consider the example of studying the effect of obesity on diabetes. First, we decide on a baseline. The baseline can be a particular date (say, January 1, 2005) or an event of interest such as patients reaching a particular age or developing a particular condition. At baseline, we take a cross-section of our population. Some of our patients already have diabetes, while others do not; some of them are obese and some are not. Since our interest is the risk of *developing* diabetes, we exclude all patients who are already diabetic; the remaining patients form our study cohort. Our exposure of interest is obesity. The obese patients form the *exposed* group and the nonobese patients the *unexposed* group. We then follow the cohort forward in time and observe how many patients develop diabetes and how many remain nondiabetic among both the exposed and the unexposed patients. We can then compute the odds (or hazard ratio) of developing diabetes among those who are obese versus those who are not.

Cohort studies are considered the best study designs for ascertaining the incidence and natural progression of a disorder, as the temporal relationship between the exposure and the outcome (Walline 2001). They are also useful in analyzing multiple outcomes that might arise after a single exposure. For example, smoking (i.e., exposure) might lead to multiple outcomes such as stroke, oral cancer, and heart disease.

However, such study designs come with certain caveats. First, selection bias is inherent in such cohort studies (Sackett 1979). For example, in a cohort study analyzing the effects of smoking on T2DM, those who smoke would differ in other important ways (lifestyle) from those who do not smoke. In order to validate the effect of exposure (i.e., smoking), both the cases and controls must be similar in all respects except for the absence/presence of exposure and the outcomes. Second, loss of subjects due to censoring might be a possibility, not only when the study is short, but also, particularly, with longitudinal studies that continue for decades. For example, progression from T2DM to associated complications such as peripheral vascular disease (PVD) and ischemic heart disease (IHD) takes around 5 to 10 years, and subjects may drop out over this long period.

6.2 Case-Control Studies

Case-control studies are study designs that look backward; that is, the study cohort is defined at a particular time (e.g., first heart attack) and the study looks backward in time to analyze the patients' exposure(s). In such studies, two groups are compared, one consisting of patients with the outcome in question (*cases*) and another one consisting of patients without the outcome (*controls*). Case-control study design can be used to identify risk factors that may contribute to an outcome by looking backward in time and comparing the exposure histories of patients across the case and control groups. Specifically, exposures that are more prominent in the case group can be risk

factors, while exposures that are more prominent in the control group can be protective factors, assuming that the subjects are similar except for the exposure.

Let us return to our example (as illustrated in Section 6.1) of estimating the effect of obesity on incident (newly developed) diabetes. We would select a group of patients (cases) newly diagnosed with diabetes and match them with another group of nondiabetic patients (controls). Matching is performed on important characteristics in an attempt to reduce the differences between the groups to the exposure alone. We would track these patients backward in time for a fixed number of years to determine their exposure history (whether they were obese or not). We can then compute the odds of developing diabetes in obese patients.

Since we start with known outcomes, these study designs are very useful in investigating (1) rare outcomes and (2) slow-onset diseases such as cancer and T2DM, where it takes a long time for the disease to develop. In both situations, finding cases—patients who have the outcome (disease) in question—would be difficult prospectively, because the study would require either a long follow-up time or a very large initial sample size. In case-control studies, cases and controls are identified at the beginning of the study and are followed retrospectively. In risk prediction, case-control study designs are widely used due to their efficiency in detecting the association between risk factors (exposure) and outcome.

Conversely, case-control study designs are inefficient when the exposure rate is low, as researchers would have to analyze the entire data cohort to identify one patient who had the exposure. For example, using a case-control study design to investigate the effect of pancreatic cancer (exposure) on T2DM (outcome) would be impractical because pancreatic cancer is very rare. In such cases, the cohort study design is more effective.

Case-control designs are susceptible to biases. For example, the exposure may occur at a time when the patient is not under study, and thus an exposure may remain undetected. Furthermore, in case-control studies, selection of the control group can also bias the results of the study and therefore researchers should provide clear eligibility criteria for the outcome being studied, such as age, gender, racial makeup, or ethnicity.

6.3 Cross-Section Studies

Cross-sectional studies seek a comparison between cases and controls by collecting data at one specific point in time (Lee 1994). Such study designs differ from case-control and cohort studies in that they aim to make inferences based on data that is collected only once rather than collected at multiple time points (Mann 2003).

Returning to our example of measuring the effect of obesity on diabetes, we would take a cross-section at a well-defined point in time (e.g., January 1, 2007). At this time, some patients are diabetic and others are not, and some patients are obese and others are not. We can compute the odds of having (not developing!) diabetes among obese patients versus those who are not obese. Note that this study design does not require the exposure to precede the outcome and thus cannot discover causal relationships. For example, some of the obese patients may have already been diabetic when they became obese.

The key advantage of the cross-sectional study design is that it only requires data at a single point in time. In risk prediction, cross-sectional study designs are widely used due to their ability to detect the association between risk factors (exposure) and outcome using data at a single point in time.

6.4 Descriptive Studies

Descriptive studies are usually designed to analyze the distribution of variables, without regard to an outcome (Walline 2001; Hennekens et al. 1987). The defining characteristic is that there are no cases or controls.

At an individual level, descriptive studies include studies reporting an unusual disease or association and at a population level. They are often used for analyzing the medical state of the population for healthcare planning. For example, descriptive designs are widely used to investigate tobacco usage within a population, age group, gender, or socioeconomic class.

One type of descriptive study is the correlational study, which aims to identify associations between conditions. Such studies typically investigate how predictive one condition is of another condition. With no information about the temporal ordering of these conditions, we do not consider them as exposures or outcomes. One common use of correlation analysis is to study the relationship between a given disease and its comorbidities (the diseases that co-occur with it) or the relationships among diseases in multimorbid disease clusters. In this situation, the study is referred to as a comorbidity analysis.

For example, consider our cohort of prediabetic patients, many of whom have different conditions (e.g., high cholesterol, obesity, high blood pressure) at baseline. Using such study designs, we can estimate the prevalence of these comorbid conditions. Further, such analysis can lead to future estimation of the sequential patterns in which such diseases occur.

7 APPROACHES

Having defined some metrics of interest in Section 5 and having selected a study design from Section 6 to transform our EHR data into an analytics matrix that data mining methods can operate on, we appear ready to answer the clinical questions we set out to solve. While the data format may suggest that we can directly apply our existing data mining techniques, the data itself creates some challenges, which we described in Section 4 in detail.

In this section, we describe approaches to address the challenges related to censoring, the irregular temporal nature of the data, and confounding. Approaches are concepts and ideas that provide high-level solutions to these challenges; they are not algorithms per se. Within these high-level solutions, concrete data mining techniques can be developed. Our focus in this section is to describe these high-level ideas, and later, partly in this section as illustrative examples, but mostly in the subsequent sections, we will discuss some concrete analytics techniques that use these approaches.

Several survey articles and books have been published on the approaches we discuss in this section. Chung et al. (1991) provided a survey of statistical methods, which analyzes the duration of time until an event of interest occurs. Such models have been widely used to analyze the survival times of various events (e.g., mortality), to analyze the time until recidivism, and for many other applications. In their paper, they summarized the statistical literature on survival analysis. A textbook by Klein and Moeschberger (2005) provides a comprehensive overview of various techniques used to handle survival and censored data. Dehejia and Wahba (2002) discussed causal inference and sample selection bias in nonexperimental settings. In their paper, they discussed the use of propensity score-matching methods for noncausal studies and illustrated several methods by applying them to data from the National Supported Work experiment.

7.1 Handling Censored Data

As discussed in Section 4.1, censoring occurs when a patient's trajectory is only partially observable. For example, suppose a study is conducted to measure the impact of a diabetes-related drug on mortality rate. In such a study, let us assume that the individual withdrew from the study after following the study course for a limited duration. In such a scenario, information about the patient's vital statistics is only available until the patient was censored. Such data is common in domains such as healthcare and actuarial science.

Survival analysis is an area of statistics that deals with censored data. These approaches usually aim to answer questions such as the following: what proportion of the population will survive

past a particular time, and what characteristics influence the probability of survival? To answer, the aforementioned clinical questions, techniques are required that can handle censoring, which is frequently present in EHRs. Techniques to handle censored data can be divided into three major categories: nonparametric, semiparametric, or parametric.

Nonparametric techniques do not rely on assumptions about the shape or parameters of the distribution of time-to-event. Examples of such techniques include Kaplan-Meier estimators (Kaplan and Meier 1958) and Nelson- Aalen estimators (Cox 1992). Rihal et al. (2002) used Kaplan-Meier estimators to analyze the incidence and prognostic implications of acute renal failure in patients undergoing percutaneous coronary intervention (PCI). Dormandy et al. (2005) used Kaplan-Meier estimates in their analysis of patients who were diagnosed with T2DM and were at high risk of data and nonfatal myocardial infarction and stroke. Rossing et al. (1996) used Nelson-Aalen estimators for analyzing the predictors of mortality in insulin-dependent diabetes. Ekinci et al. (2011) used such nonparametric techniques for exploring salt intake consumption and mortality in patients diagnosed with T2DM.

Parametric techniques often rely on theoretical assumptions about the shape or parameters of the distribution of time-to-event. Examples of such techniques are the accelerated failure time models (AFT models) (Keiding et al. 1997), which are an alternative to the widely used proportional hazard models. Using such models, Babuin et al. (2008) determined whether troponin elevations influence short- and long-term mortality in medical intensive care unit patients. Wilson et al. (2008) used AFT models to predict cardiovascular risk by using predictors such as age, gender, high-density lipoprotein cholesterol, diabetes mellitus (DM), cholesterol, smoking status, systolic blood pressure, and body mass index (BMI).

Semiparametric techniques have both parametric and nonparametric components. An example of such a technique is the Cox proportional hazards model. These models have a nonparametric time-dependent baseline hazard (risk) and assume that the covariates impart a risk on the patient that is proportional to the baseline hazard. Yadav et al. (2015b) used the proportional hazards model for risk assessment of comorbid conditions in T2DM. They identified how risks vary across subpopulations for the same outcome. The subpopulations were defined by using diagnosis codes, such as hypertension, hyperlipidemia, and T2DM, with time to death being modeled as the outcome of interest. Martingale residuals were used to compute the risks.

7.2 Handling Irregular Time-Series Data

Data stored in EHRs is usually collected through a longitudinal study. In such studies, the subject outcomes, treatments, or exposures are collected at multiple follow-up times, usually at irregular intervals. For example, patients diagnosed with T2DM might be followed over time, and annual measures, such as hemoglobin A1c and GFR, are collected to characterize the disease burden and health status, respectively. As these repeated measures are correlated for a subject, they require sophisticated analysis techniques. In what follows, we describe techniques that are widely used to handle these repeated measurements. In particular, we cover marginal and conditional models, respectively. These models handle unevenly spaced (irregular) EHRs by assuming a correlation structure among multiple clinical observations of a patient recorded at different time points.

Marginal models (Robins et al. 2000), which are also known as the population-averaged models, make inferences about population averages. In such models, the target of inference is usually the population and these models are used to describe the effect of covariates on the average response. They are also used to contrast the means in subpopulations that share common covariate values. For example, consider a cohort of prediabetic patients with elevated cholesterol levels. In this cohort, if we are interested in estimating the progression of patients to full-blown T2DM, we would probably want to use the population-averaged coefficients. Generalized estimating

equations (GEEs) are most commonly used for parameter estimation in marginal models. This approach is computationally straightforward and, with care, can handle missing data, even when the covariance has been misspecified. Such modeling techniques are widely used in epidemiological studies, particularly in multisite cohort studies, as they incorporate the effect of unmeasured dependence across outcomes.

Conditional models (Laird and Ware 1982), which are also known as the locally averaged models, make inferences about individual subjects. The estimates are based on averaging or smoothing done by the model, but more locally, they are based on sources of dependence in estimating model parameters. For example, consider once again our aforementioned cohort of prediabetic patients with elevated cholesterol levels. In this cohort, if we are interested in estimating the effect of statin across every individual, we would use conditional models.

7.3 Handling Confounding

A confounding variable can be defined as an extraneous variable that correlates with both the dependent and the independent variable. To handle confounding, we discuss techniques such as propensity scoring and inverse probability weighing.

Statistical matching techniques such as propensity score matching (PSM) (Peikes et al. 2008) aim to estimate the effect of an intervention by incorporating the effect of covariates that predict how likely it is to receive the intervention. They aim to reduce the bias caused by confounding variables. PSM creates a group by employing the predicted probability of group membership, which is usually obtained from logistic regression. The key advantage of PSM is that by using a linear combination of features, it balances both the intervention and the nonintervention group on a large number of covariates. One disadvantage of PSM is that it only accounts for known covariates, that is, variables that are observed. Another issue is that PSM requires large samples, with substantial similarities in terms of subjects between treatment and control groups. Polkinghorne et al. (2004) used PSM to analyze the inception and intervention rate of native arteriovenous fistula (AVF).

Inverse probability weighting (Hogan and Lancaster 2004) is a statistical technique for calculating statistics that aims to standardize results to a target population of interest as compared to the population from which the data was collected. The motivation is that factors, such as cost, time, or ethical concerns, often prohibit researchers from directly sampling from the target population. Instead of adjusting for the propensity score, inverse probability weighting assigns a weight to each subject, and this weight is used in further statistical computations. Robinson et al. (2011) used inverse probability weighting for examining whether lower serum levels of serum 25-hydroxyvitamin are associated with increased risk of developing type 2 diabetes.

8 CLINICAL DATA MINING METHODOLOGIES

The discipline of EHR data mining stands at the intersection of epidemiology, biostatistics, and general data mining. From epidemiology and biostatistics, clinical data mining has borrowed study design, the methodology that allows us to organize EHR data into a matrix that is amenable to the application of data mining algorithms and can answer meaningful clinical questions correctly. It has also borrowed basic approaches from biostatistics and epidemiology to address the challenges that EHR data poses, including censoring, analysis of irregular time-series data, and methodologies for causal inference.[1] With the success that data mining has achieved in many commercial applications, the expectation from data mining in the health care domain is high and its contributions are becoming apparent. In this section, we review these contributions.

[1]The roots of causal inference are in computer science, but it has been embraced by epidemiology and biostatistics, resulting in the development of the advanced techniques we described earlier.

Table 1. Structure for Section 8

	Unsupervised	Supervised
Time Agnostic	Atemporal Descriptive Studies (8.1.1)	Cross-Sectional Studies (8.2), Cohort and Case-Control Studies (8.3.1)
Time Aware	Temporal Descriptive Studies (8.1.2)	Cohort and Case-Control Studies (8.3.2)

Traditionally, data mining techniques (Tan et al. 2006) are broadly categorized as supervised or unsupervised: supervised methods take an outcome into account, while unsupervised methods simply learn from the structure of the data. This distinction neatly maps to study designs: supervised techniques are applicable to analytical studies and unsupervised techniques to descriptive studies. The hallmark of EHR data is its temporal nature, suggesting that data mining techniques be further categorized based on their ability to take time into account. We call a data mining algorithm and its resulting model *time aware* if the outcome of interest depends on time; we call it *time agnostic* if it builds a model that does not take time into account.

Although EHR data is inherently temporal, time is not always of relevance. The clinical question we aim to answer may be *temporal* if time is of relevance (i.e., time is part of the question) or it may be *atemporal* (not temporal) if time is not part of the question. Atemporal questions are naturally answered by time-agnostic data mining techniques. On the other hand, temporal questions either can be answered by time-aware models or, if the question can be transformed into a simpler atemporal question, can be solved using time-agnostic models. For example, predicting the risk of 30-day mortality after surgery is a temporal question (time is part of the question), but it can be solved using time-aware models (e.g., Cox model) or time-agnostic models (e.g., logistic regression).

The study design dictates whether a question can be temporal or atemporal, and it also determines in large part whether any of the challenges posed by EHR data can be successfully addressed. For this reason, we describe data mining techniques that are commonly applied in the context of the applicable study designs. In Table 1, we present the structure of the following subsections.

8.1 Descriptive Studies

Descriptive studies represent the broadest variety of inquires we can undertake, ranging from simple statistics (prevalence rate, incidence rate) to descriptions of the progression of a particular disease via case studies. Such simple applications do not require data mining, but data mining techniques enable more advanced applications including comorbidity analysis and trajectory mining. While descriptive studies cover a wide range of applications, their defining characteristic is that exposure and outcome are not considered. Without a particular outcome, we cannot have outcome labels and hence the problem at hand is *unsupervised*.

Descriptive studies are commonly used to answer both temporal and atemporal clinical questions. For example, estimating prevalence rates at a particular time is an atemporal clinical question, while extracting the trajectory of a patient as sequences of diagnosis codes is naturally a temporal clinical question. Therefore, both time-aware and time-agnostic data mining techniques are applicable for descriptive studies.

8.1.1 Atemporal Descriptive Studies. Atemporal descriptive studies can arguably be handled using standard textbook methods. A prototypical application of this nature would be to take a snapshot of the population at a particular time and cluster the patients based on the conditions they present.

In this section, we discuss how unsupervised techniques have been widely used for *identifying clusters* of patients who have similar characteristics (e.g., demographics, medications, diagnosis codes, laboratory test results) and for finding *associations* between clinical concepts (e.g., medications, diagnosis codes, and demographic attributes). Next, we describe how these techniques make use of the approaches we discussed in Section 7.

Clustering: Gotz et al. (2011) used clustering techniques for identifying a cohort of patients similar to a patient under observation. They used the cohort as a surrogate for near-term physiological assessment of the target patient. Roque et al. (2011) stratified patients using hierarchical clustering, where the distance between patient records was computed using the cosine similarity of diagnosis codes. Along similar lines, Doshi-Velez et al. (2014) investigated the patterns of co-occurring diseases for patients diagnosed with autism spectrum disorders (ASD). They identified multiple ASD-related patterns using hierarchical clustering. They further discussed how the aforementioned patterns can be attributed to genetic and environmental factors. Kalankesh et al. (2013) noted that representing the medical state of a patient with diagnosis codes can lead to sparse clusters since EHRs often contain a large number (i.e., thousands) of diagnosis codes. To overcome this problem, they used Principal Component Analysis (PCA) (Dunteman 1989) to reduce the dimensionality, thereby making the structure more amenable for visualization and clustering.

Association Analysis: Association rule mining techniques (Agrawal et al. 1994) such as Apriori have also been used on EHR data to identify associations among clinical concepts (medications, laboratory results, and problem diagnoses). These techniques have the ability to discover associations and interpretable patterns from EHR data. However, the performance of such techniques often deteriorates when there are a large number of clinical variables present in EHRs. Wright et al. (2010) used the Apriori framework to detect transitive associations between laboratory test results and diagnosis codes and between laboratory test results and medications. For example, they observed some unexpected associations between hypertension and insulin. They attributed this finding to co-occurring diseases and proposed a novel way to identify such transitive associations. Cao et al. (2005) used co-occurrence statistics to identify direct and indirect associations among medical concepts. Holmes et al. (2011) used statistical approaches to detect associations between rare diseases. They observed that analyzing cohorts composed of sick patients leads to identification of significant findings. Shin et al. (2010) used association rule mining to identify comorbidities (e.g., non-insulin-dependent diabetes mellitus (NIDDM) and cerebral infarction) that are strongly associated with hypertension. Hanauer et al. (2009) used statistical tests to observe common pathways for diseases such as granuloma annulare and osteoarthritis.

Challenges and Limitations: In Section 4, we identified a number of challenges, centered around three key issues: censoring, irregular time series, and causation. In this paragraph, we describe the limitations of these methods in relation to these challenges.

While individual techniques all have their own limitations, methods that are applied specifically to atemporal descriptive studies share some limitations stemming from the nature of the study design. Atemporal descriptive studies, as with all atemporal studies, deal with information collected at one time instant (at *baseline*) or summarized until baseline. With only one time point, this study design completely sidesteps the challenges stemming from irregular time series, but at the cost of losing information: a potentially long sequence of information has been aggregated to a single time point.

Atemporal studies (and the data mining methods that are applicable to these studies) are susceptible to biases stemming from censoring. Right censoring is obviously present as we cannot observe any change in the patients' condition after baseline. Suppose we have two patients, neither

of whom has a condition at baseline, but one develops it shortly after baseline, while the other develops it only decades later (or not at all). These patients are clearly different for the purpose of the analysis, but the atemporal methodologies consider them identical, which constitutes information loss. A similar situation arises with left censoring: we may not know how long a patient has had a condition, but if we had that information, we could obtain more accurate results.

Causal inference is generally not possible under atemporal descriptive studies. First, descriptive studies do not differentiate between exposures and outcomes; thus, trying to establish causation is meaningless. Second, even if we attempted to establish causation between two conditions, due to the atemporal nature of the study, we cannot even ascertain that one happens before the other.

While these methods are rather limited, their popularity stems from their simplicity. Most off-the-shelf data mining algorithms are directly applicable.

8.1.2 Temporal Descriptive Studies. Time plays an important role in clinical questions, which, for instance, could be concerned with the sequence of events or timing between events. Standard textbook data mining techniques exist to solve such problems (e.g., sequence mining, Markov models), but to achieve better results, significant improvements have been proposed. We broadly classify the approaches that can be carried out using such techniques as those which use time-aware techniques (e.g., sequence mining, time-lagged correlations, etc.) and those which simplify the problem and apply time-agnostic techniques (e.g., temporal-abstraction (summarizing the longitudinal data) and Hidden Markov Model (HMM) trajectory clustering (using HMMs to simplify away time so that standard clustering is applicable)).

Temporal Abstraction Framework: The temporal abstraction framework has been frequently used to prepare patterns from EHR data. Patterns can be abstracted using state representations (e.g., high, medium, or low) or trend representations (e.g., increasing, decreasing, constant). Shahar (1997) provided a mechanism to abstract patterns from unevenly spaced time series. Such time series are common in EHR data elements, such as laboratory test results and vitals. They further proposed temporal logic relations to combine patterns generated from univariate time series. Sacchi et al. (2007) extended the temporal abstraction framework to generate temporal association rules (TARs). In TARs, the antecedent and the consequent both consist of temporal patterns generated using the temporal abstraction framework. Jin et al. (2008) further extended the TAR framework to generate rules for mining surprising patterns. In such patterns, certain events led to unexpected outcomes—for example, taking multiple medicines together sometimes causes an adverse reaction. Batal et al. (2009) used the temporal abstraction framework to propose the Segmented Time Series Feature mining algorithm for identifying the frequent patterns from an unevenly sampled time series. Such modeling techniques have their own set of challenges. Patterns generated from individual patient time series are susceptible to noise. Further, such patterns can be of uneven temporal duration. Marlin et al. (2012) developed a probabilistic clustering method along with temporal abstraction to mitigate the effects of unevenly spaced data, which is inherent in EHRs.

Trajectory Clustering: Clustering techniques have also been used to group EHR data. Ghassempour et al. (2014) used HMMs to cluster patient medical trajectories. In their approach, they used both categorical variables (diagnosis codes) and continuous variables (vitals and laboratory test results) for clustering. They first mapped each medical trajectory to an HMM and then used KL divergence to compute the distance between two HMMs.

Sequential Rule Mining: Researchers have explored sequential association rule mining techniques for identifying causal relationships between diagnosis codes. Hanauer and Ramakrishnan (2013) identified pairs of ICD-9 codes that are highly associated. They observed interesting temporal relationships between hypothyroidism and shingles (herpes reactivation). Liao and Chen

(2013) proposed a sequential pattern mining approach to mine sequences with gap constraints. Such gaps represent the delay between two concepts. Hripcsak (2013) measured lagged linear correlation between EHR variables and healthcare process events. In their analysis, they considered five common healthcare process events—inpatient admission, outpatient visit, inpatient discharge, ambulatory surgery, and emergency department visit—and computed their correlation with several EHR variables such as laboratory values and concepts extracted from clinical notes.

Challenges and Limitations: Methods applied in the context of temporal descriptive studies have access to granular temporal information, and thus, in theory, they can utilize more information than the atemporal descriptive methods from Section 8.1.1. Even when the methods themselves are unable to utilize the granular temporal information, the study design allows them to produce results that are qualitatively different. While atemporal descriptive studies are limited to using prevalence (proportion of patients in a population presenting with a condition), temporal descriptive studies can use incidence (proportion of patients *developing* disease) as well as prevalence.

Naturally, these techniques can overcome the challenges of irregular time series either directly (e.g., time-lagged correlation) or through transforming a temporal problem to an atemporal one by temporal abstraction or by enumerating the sequences of the events.

Another challenge that temporal studies can overcome is handling censored data. The information loss as a result of censoring that we described in the previous section can be reduced. In case of atemporal descriptive studies, we have absolutely no information regarding when patients develop each condition of interest. Instead, the only information we have is a binary indication of whether they have it at baseline. In contrast, in case of temporal descriptive studies, even in the presence of left censoring, we know that a patient has had the condition for at least a certain amount of time, and conversely, in case of right censoring, we know that the patient remains free of that condition for at least a certain amount of time. While the techniques we described in this section did not make use of this information, it is possible to develop techniques based on survival analysis that can.

As with atemporal descriptive studies, causation remains impractical for temporal descriptive studies. In case of causation, we are generally interested in the causal effect of an exposure on an outcome. These methods are still operating under the framework of descriptive studies, and thus they do not distinguish between exposure and outcome.

8.2 Cross-Sectional Design

Cross-sectional study design, like many descriptive studies, also considers a patient population only at a single point in time. The key difference is that unlike descriptive studies, cross-sectional studies have outcomes of interest. The aim of such studies is usually to identify and analyze the associations between risk factors and the outcome of interest. In such studies, data is often collected on individual characteristics, such as exposure to risk factors, demographic attributes, and information about the outcome. In what follows, we will describe the techniques often used for such study designs along with examples of research carried out using data mining algorithms.

When a study is designed as cross-sectional, supervised time-agnostic data mining techniques are the natural modeling choices. Since interest in a specific outcome is very natural and there is great appeal in solving these problems through relatively simple supervised nontemporal data mining techniques, this class of techniques has been applied to a broad spectrum of problems, including risk prediction for hospitalization, rehospitalization, and diagnostic and prognostic reasoning.

Rule-Based Methodologies: White et al. (2013) conducted a large-scale study for analyzing web search logs for detection of adverse events related to the drug pair paroxetine and pravastatin.

They analyzed whether the drug interaction leads to hyperglycemia. Iyer et al. (2014) used NLP techniques for mining clinical notes to identify events related to adverse drug-drug associations. Haerian et al. (2012) hypothesized that adverse events might be caused by the patient's underlying medical condition. Along similar lines, Vilar et al. (2012) used disproportionality-based techniques to analyze adverse drug events related to pancreatitis, Li et al. (2014) used penalized logistic regression to analyze associations between ADRs, and Epstein et al. (2013) used NLP techniques to analyze medication and food allergies. Supervised nontemporal methodologies have been frequently used in the form of rule-based techniques for cohort identification. Phenotyping algorithms for diseases, such as celiac disease, neuropsychiatric disorders, drug-induced liver injury, and T2DM (Pathak et al. 2013; Carroll et al. 2011; Xu et al. 2011), have been widely explored. Supervised pattern mining approaches using the temporal abstraction framework have been used for predicting heparin-induced thrombocytopenia (HIT) (Batal et al. 2009). Batal and Hauskrecht (2010) used such methodologies to generate minimal predictive rules for Heparin Platelet Factor 4 antibody (HPF4) test orders. They further extended their approach by introducing the minimal predictive pattern (MPP) framework wherein they directly mine a set of highly discriminative patterns (Batal et al. 2012). Those patterns were later used for classification of related tasks.

Bayesian Networks: Bayesian networks have also been used to model EHRs for diagnostic reasoning (reconstructing the medical state of the patient using laboratory test results), prognostic reasoning (prediction about the future), and discovering functional static interactions between the outcome and the predictors (Lucas et al. 2004). Zhao et al. (2011) integrated EHR data with knowledge from other sources such as Pubmed to develop a weighted Bayesian network for pancreatic cancer prediction. They also discussed how their approach can be used to detect clinically irrelevant variables for disease prediction. Sverchkov et al. (2012) compared clinical datasets by capturing clinical relationships between individual datasets by using Bayesian networks. The multivariate probability distributions were then used to compare the clinical datasets.

Challenges and Limitations: Methods under cross-sectional design are similar to methods under descriptive atemporal design in that they are all atemporal: they only consider a cross-section taken at one point in time. Therefore, the limitations associated with atemporal design, namely, the lack of ability to handle censoring and to make use of the granular time information in the EHR, apply to cross-sectional designs as well.

The key difference between a cross-sectional and descriptive study design lies in the existence of comparison groups and thus a distinction between exposure and outcome. Although this difference allows for qualitatively different results—we can now measure risks—causal inference is not practical because we cannot establish the temporal relationship between exposure and outcome. To establish causation, we need to ascertain that the exposure precedes the outcome.

8.3 Cohort and Case-Control Study Design

These two study designs, cohort and case control, add a temporal dimension to cross-sectional design; that is, they compare patient groups with different exposures over time and record their outcomes. They differ in the direction in which time is observed: in cohort studies, patients are followed from exposure to outcome, while in case-control studies, patients are followed backward from outcome to exposures. Although this difference has far-reaching consequences on the required sample sizes, exposure rates, and the metrics we can estimate, once the design matrix has been constructed, the same data mining methods apply to both of these study designs. Hence, we consider these two designs together.

What is common across these study designs is that they are best suited to answer *temporal* questions. As is typical with temporal questions, we can either use time-aware models or simplify

the question such that it can be answered using time-agnostic models. In the following paragraphs, we provide examples of both.

8.3.1 Time-Agnostic Models for Cohort and Case-Control Studies. Supervised time-agnostic models are commonly employed when time-to-event can be removed from the clinical question. For example, time-to-rehospitalization can be simplified to the binary outcome of 30-day rehospitlaization (yes/no), which does not include time, or we can consider in-hospital or ICU mortality (Herasevich et al. 2013), where the time dimension is removed by considering the entire duration of the hospitalization or the stay in the ICU.

Applications of supervised time-agnostic modeling mainly focus on risk prediction, such as predicting the onset of neonatal sepsis (Mani et al. 2014); the risk of depression (Huang et al. 2014), dementia (Maroco et al. 2011), T2DM (Mani et al. 2012), and retinopathy in patients suffering from type 1 diabetes mellitus (T1DM) (Skevofilakas et al. 2010); potentially preventable events (Sarkar and Srivastava 2013); or 30-day hospital readmissions (Cholleti et al. 2012; Park and Ghosh 2014); or when we look at the "risk" of a positive outcome such as survival of heart-lung transplant (Oztekin et al. 2009) and breast cancer (Sarvestani et al. 2010) patients.

Risk prediction also includes comparative risk assessment, such as comparing cancer survival rates across continents (Coleman et al. 2008), exploring relationships between hospital surgical volumes and the 5-year relationship of stomach cancers (Nomura et al. 2003), comparing dosage volumes of warfarin in European Americans and African Americans (Ramirez et al. 2012), and comparing postpartum depression rates in Asian American subgroups (Indian, Vietnamese, Chinese, Korean, Filipino, Japanese) (Goyal et al. 2012).

For these analyses, almost all flavors of common predictive modeling techniques have been applied: decision trees (Mani et al. 2012; Sarvestani et al. 2010; Austin et al. 2012); ensemble techniques including bagging, boosting, and random forests (Cholleti et al. 2012; Kawaler et al. 2012; Mani et al. 2014; Karnik et al. 2012); naive Bayes (Kawaler et al. 2012; Karnik et al. 2012; Sarvestani et al. 2010); support vector machines (Mani et al. 2012); and logistic regression (Zhai et al. 2014; Mani et al. 2012; Cholleti et al. 2012; Huang et al. 2014; Chang et al. 2011).

Besides risk prediction, time-agnostic supervised methods can be used for supervised feature creation (Aliferis et al. 2010) and for clinical questions that are more complex than risk assessment. Ghalwash and Obradovic (2014) proposed a predictive modeling technique to find a suitable duration of the hemoadsorption (HA) therapy control and observed that their method led to substantial monetary savings. Sun et al. (2014) worked on predicting the risk and timing of deterioration in hypertension control by analyzing critical points in time at which hypertension status is at borderline (clinical limit separating in control vs. out of control). Wang et al. (2015b) developed a dynamic Poisson autoregressive model for flu forecasting wherein they allowed the autoregressive model to change over time. Panahiazar et al. (2015) built a heart failure risk prediction model using several machine-learning techniques that addressed multiple comorbidities, which led to an improvement in prognostic predictive accuracy. Wang et al. (2014a) proposed multilinear sparse logistic regression to handle data in the form of multidimensional arrays. They used their methods to predict the onset risk of patients with Alzheimer's risk and heart failure.

Challenges and Limitations: Case-control and cohort study designs are the most flexible and informative designs among the study designs we consider. These designs have dedicated exposures and outcomes, which allows the estimation of risks. They also have detailed temporal information, which allows the measurement of the time-to-event and establishes the precedence (temporal ordering) of events. Unfortunately, time-agnostic methods handicap the study design. Simplifying this study design to become atemporal effectively renders the study design into a cross-sectional design with some amount of extra information about outcome (did it happen within a certain

time-frame?). Since the design has been effectively reduced to cross-sectional, the limitations of cross-section designs stand, but to a lesser degree.

In case of a cross-sectional design, we have no information about the timing of the outcome. In contrast, in case of case-control studies, even when we use time-agnostic methods, we usually have some information about the time of the outcome; for example, the patient did not develop the disease in a certain timeframe (say, 5 years). The information these methods discard concerns patients who did not develop the disease. Specifically, patients who were censored (lost to follow-up within 5 years without developing the disease) are ignored, even though these patients carry partial information, namely, that they did not develop the disease for some amount of time (<5 years).

The key advantage of these methods over cross-sectional design lies in causation. Although we do not have detailed time-to-event data, we know that the exposure precedes the outcome, and we also have a distinction between exposure and outcome. Thus, causal inference is possible.

8.3.2 Time-Aware Models for Cohort and Case-Control Studies. Supervised time-aware models are utilized when the clinical question cannot be simplified or if the simplification to time-agnostic modeling comes at a significant loss of information. Such questions focus on the time-to-event itself (clearly cannot be simplified), sequences of events, or situations where time-to-event carries additional information about the outcome. Continuing with the example of 30-day rehospitalization, by simplifying the outcome to binary yes/no, we lose information since we ignore whether the patient was rehospitalized in (say) 7 days versus 20 days. The former case is clearly more severe.

Many of the temporal clinical questions are related to right censoring. Survival modeling, which was specifically developed for this purpose, is the quintessential technique for this study design. Survival modeling is a suite of techniques with various specializations that share the common characteristic of being able to handle time and censoring. Other techniques that incorporate the effect of time include dynamic Bayesian networks, sequential pattern mining, and deep learning.

Survival Modeling: Wells et al. (2008) hypothesized that patients diagnosed with T2DM have an increased risk of mortality. They used Cox proportional hazards regression with time to death as the outcome. They also observed that certain interaction terms involving medications and age were significant indicators. Vinzamuri and Reddy (2013) extended Cox proportional hazards regression to effectively capture the grouping and correlation of features. They proposed novel regularization frameworks to handle the correlation and sparsity present in EHR data. They demonstrated the applicability of their technique by identifying clinically relevant variables related to heart failure readmission. Vinzamuri et al. (2014) proposed a novel active-learning-based survival model wherein continuous feedback from a domain expert can be utilized to refine the model.

Survival modeling techniques on time-to-event data have been explored widely in the past. Cox regression (Cox 1992; Vinzamuri and Reddy 2013) is one of the most commonly used survival regression models. Its formulation, namely, its semiparametric nature, with the mild assumption of the proportionality of hazards, makes it ideal for many practical applications in fields such as economics (Wooldridge 1992), healthcare (Ikeda et al. 1991; Liang et al. 1990; Lumley et al. 2002), and recommendation systems (Sharma et al. 2015; Anastasiu et al. 2016).

Cox models, as with most other regression techniques, are susceptible to overfitting. Standard regularization techniques, developed for other regression methods, have been applied to Cox models as well. Lasso (Tibshirani et al. 1997) and elastic-net regularized Cox models (Simon et al. 2011b) have been developed and have been further extended by regularizing them with convex combinations of L1 and L2 penalties (Zhang and Lu 2007). We are not aware of regularization for time-dependent covariate Cox models (Therneau and Crowson 2014), which would be a straightforward extension.

Reddy et al. (Vinzamuri et al. 2014) proposed a new survival modeling algorithm that uses a sophisticated discriminative gradient-based sampling scheme and observed better sampling rates as compared to other sampling strategies. To handle correlated and grouped features, they proposed correlation-based regularizers with Cox regression, which are commonly seen in many practical problems (Vinzamuri and Reddy 2013). Kuang et al. (Zhang et al. 2013) proposed Net-Cox, a network-based Cox regression model to handle the high dimensionality of high-throughput genomic data. They further applied their model to a large-scale survival analysis across multiple ovarian cancer datasets.

Support vector machine (Hearst et al. 1998) models have also been extended to handle censored data (Khan and Zubek 2008; Evers and Messow 2008; Shivaswamy et al. 2007; Van Bellev et al. 2007; Shiao and Cherkassky 2014). In such techniques, often the task is converted into a ranking problem via the concordance index. This in turn is efficiently solved using convex optimization techniques. Along similar lines, Khosla et al. (2010) proposed algorithms that combine margin-based classifiers along with censored regression algorithms to achieve higher accuracies (concordance in this case). They used their technique to identify potentially novel risk markers for cardiac problems.

Research has also been carried out on extending decision trees to handle censored data (Gordon and Olshen 1985). Ishwaran et al. (2008) proposed Random Survival Forests for analyzing right-censored survival data. They analyzed splitting rules for growing survival trees, introduced a new measure of mortality, and applied it for patients diagnosed with coronary artery disease. Neural nets have also been adapted to handle censored data with varying results (Kattan et al. 1998; Snow et al. 1994). Forensic style analysis with survival trajectories (Yadav et al. 2015a) have also been explored.

Dynamic Bayes Networks: While survival models are by far the predominant type of models, other methods that can incorporate temporal information also exist. Dynamic Bayesian networks (DBNs) (Melnyk et al. 2013) have been used to model temporal relationships among EHR variables (Rana et al. 2015). Nachimuthu et al. (2010) used DBNs to model temporal relationships between insulin and glucose homeostasis. The modeling was further used to predict the future glucose levels of a patient admitted in an ICU. They also discussed the reasons for using first-order Markov models to model the temporal relationships. For modeling organ failure in patients admitted to an ICU, Sandri et al. (2014) used DBNs with multiple order dependencies to impose restrictions on the causal structure. In their model, each timestamp represented a day. They further imposed several constraints, for example, that no patient discharges were recorded on the second day and that all patients were either deceased or considered discharged on their seventh day. Such constraints were imposed to reduce the complexity of the model. Along similar lines, Rose et al. (2005) used DBNs to assist physicians in monitoring the weight of patients suffering from chronic renal failure, Gatti et al. (2011) used it to model heart failure, and Peelen et al. (2010) used hierarchical DBNs for modeling organ failure. Expectation-Maximization was used to learn conditional probabilities for these DBNs.

Sequential Pattern Mining: In the realm of supervised temporal pattern mining, research has extended the temporal abstraction framework by mining recent temporal patterns for monitoring and event detection problems in patients suffering from T2DM (Batal et al. 2012). Sengupta and Naik (2013) used similar techniques for detecting sequential rules associated with the early identification of brain tumors. Simon et al. (2013) proposed survival association rule mining (SARM) techniques, which use survival modeling techniques to incorporate the effects of dosage and other confounders, such as age and gender.

Deep Learning: Deep learning has been widely used in conjunction with EHRs for identification of novel phenotypes and robust clinical decision support systems. Such techniques have been applied to predict the onset of T2DM and congestive heart failure, as well as for improving sleep. It is believed that information extraction from such robust techniques can be used to improve healthcare outcomes and delivery. Shickel et al. (2017) provided a comprehensive review of recent research performed in EHRs using deep learning techniques. In their survey, they illustrate how deep learning techniques have been used for information extraction (medical concept extraction, temporal event extraction, relation extraction, and abbreviation extraction), representation learning (medical concept representation, patient health state representation), outcome prediction (static event prediction, dynamic event prediction), phenotype discovery (novel phenotype discovery, updating of existing definitions), and clinical text de-identification.

In particular, Lasko et al. (2013) used deep learning for phenotype discovery in clinical data. In their analysis, they used a deep learning architecture in conjunction with Gaussian process regression to generate phenotypic features that identified multiple population subtypes, thereby distinguishing the uric-acid signatures of acute leukemia versus gout. They further observed that the phenotypic features were as accurate as gold standard features created by domain experts. Liang et al. (2014) hypothesized that creating efficient feature representations requires massive manual efforts. To overcome this, they used deep-learning-based architectures, which can express different complex concept levels with multiple layer networks, for unsupervised feature extraction. Extracted features were then used to perform supervised learning via SVMs. Miotto et al. (2016) proposed a novel unsupervised feature learning algorithm using a three-layer stack of autoencoders that facilitated predictive modeling for various diseases such as T2DM, cancer, and schizophrenia. They observed increased predictive performance when compared with raw EHR data and traditional feature engineering techniques. Choi et al. (2015) used Recurrent Neural Networks to predict the medication and diagnosis classes for the next visit using longitudinal data consisting of 260K patients. They further validated their study using another cohort and observed how deep-learning-based architectures can be used to achieve better accuracy for noisy and missing clinical data.

To overcome the sparsity, noise, and temporality associated with EHRs, Cheng et al. (2016) used deep learning networks for discovering patient phenotypes. In their proposed methodology, they first converted EHR data into a matrix with time being one of the dimensions and the sequence of events being the other dimension. Such matrices are then fed into a four-layer convolutional neural network model. The first layer consists of the EHR matrices, the second layer consists of the phenotypes, the third layer imposes sparsity on these phenotypes in order to retain only significant phenotypes, and the fourth layer is a soft-max layer for predicting the disease of interest. Further, multiple fusion techniques were used to handle the temporal smoothing of the patient EHRs. Along similar lines, Che et al. (2017), motivated by the success of stacked autoencodes along with recurrent neural networks, used multilayer convolutional networks to analyze longitudinal, nonlinear EHR data. They hypothesized that their model can learn short- and long-term linear dependencies, thereby discovering novel phenotypes that can then be used for the development of individualized and personalized treatment plans. They demonstrated successful results in risk prediction for patients diagnosed with T2DM and congestive heart failure. Ranganath et al. (2016) trained a Long Short-Term Memory (LSTM) recurrent neural network to model longitudinal EHR data consisting of laboratory test results. They also discussed how their technique performs better than logistic regression with hand-crafted features and observed that representation learning significantly outperforms the baseline.

Challenges and Limitations: These techniques are by far the most successful in terms of overcoming EHR-related challenges. Right-, left-, and interval-based censoring can be easily handled

by employing techniques such as Cox proportional hazards regression and accelerated failure models. The biggest advantage of this group of techniques is their ability to handle causation. As these techniques have comparison groups (i.e., case and control) and can handle time-to-event data, causal analysis can be readily performed. Further, causation, adjusting for measured confounders, can also be analyzed by using marginal structural models and structured nested models. However, the literature of such techniques in computer science is very sparse. One area where more work is needed is in handling unmeasured confounders for the disease of interest. Similarly, more research is required in areas where the effects of confounders need to be adjusted for time-to-event data.

9 DISCUSSION

Despite its infancy, the healthcare data mining literature is very rich. Table 2 provides a succinct representation of the major research carried out using data mining techniques in conjunction with EHRs. For every major application area, presented in Section 2, the table lists publications related to the various methodologies from Section 8. In the rest of this section, we explore, discuss, and present novel insights on how data mining techniques have been utilized for EHRs. The first three subsections correspond to three different viewpoints from which Table 2 can be viewed, namely, the application, the study design, and the data mining methodology. In the fourth and last subsection, we discuss what we believe is the most important barrier to the wide-spread use of data mining in clinical practice.

9.1 Applications

Risk prediction is a popular application because it is the most natural and immediately impactful application. With numerous data mining and statistical tools and techniques readily available, such analyses can be readily performed. The literature using such analyses is rich, providing researchers with previous results to which their findings can be compared.

Another popular application area is understanding the natural history of the disease. Again, understanding the prevalence, incidence, and coincidence of diseases is the foundation on which policy decisions can be made, and thus disciplines like epidemiology have spent considerable effort on this application. Also, off-the-shelf statistical or data mining software makes these studies accessible to a wide range of researchers.

Conversely, certain application areas are virtually unexplored. We found three major reasons for this: (1) these applications may be technically difficult, requiring knowledge of concepts that are not commonly known in data mining; (2) they require extensive collaborations; or (3) the applications in question are simply not practical.

Quantifying the effect of interventions embodies the first reason. It is technically difficult since it requires two key technical elements that are not usually part of the standard data mining toolbox: study design and causal inference. On the surface, quantifying the effect of interventions, say, the effect of a drug on an outcome, seems to be a simple causal inference problem like those we described earlier in the survey, and thus, some of the solutions, such as propensity score matching, appear to be directly applicable. However, a number of epidemiological considerations are necessary to arrive at a valid conclusion. First, we need to consider the comparison groups. In a population, there are two groups of patients: those who are subject to the intervention and those who are not. Clearly, patients who are subject to the intervention form our *exposed* group, but who are the controls (the *unexposed* group)? Patients who are not subject to the intervention are of many kinds: patients with no intervention at all; patients who require a weaker intervention; patients who are subject to an alternative, equally effective intervention; and patients who have already progressed and require a stronger intervention. These groups could all serve as controls, but depending on which group we choose, we answer a different clinical question. For example, if

Table 2. Data Mining Research in EHRs

Understanding the Natural History of Disease	
Descriptive Atemporal	Dunteman (1989), Zhang and Zhang (2002), Cao et al. (2005), Holmes et al. (2011), Zhang et al. (2014), Hanauer et al. (2009, 2013), Gotz (2016), Gotz and Stavropoulos (2014), West et al. (2015), and Monsen et al. (2010, 2015)
Descriptive Temporal	Hanauer and Ramakrishnan (2013), Liao and Chen (2013), Hripcsak (2013), Sacchi et al. (2007), Jin et al. (2008), Batal et al. (2009), Patnaik et al. (2011), and Munson et al. (2014)
Cross-Sectional	Dasgupta and Chawla (2014), Albers and Hripcsak (2010, 2012)
Time-Agnostic Case Control or Cohort	
Time-Aware Case Control or Cohort	Nachimuthu et al. (2010), Verduijn et al. (2007), Batal et al. (2012), Choi et al. (2016), and Perotte and Hripcsak (2013)
Cohort Identification	
Descriptive Atemporal	Roque et al. (2011) and Bauer-Mehren et al. (2013)
Descriptive Temporal	Gotz et al. (2014), Schulam et al. (2015), Lasko et al. (2013), and Wang et al. (2015a)
Cross-Sectional	Wang et al. (2013b), Peissig et al. (2012), Pathak et al. (2012), Nadkarni et al. (2014), Chen et al. (2015, 2016), Hripcsak et al. (1997), and Hripcsak and Albers (2013)
Time-Agnostic Case Control or Cohort	Carroll et al. (2011), Xu et al. (2011), and Sarvestani et al. (2010)
Time-Aware Case Control or Cohort	Albers et al. (2014), Liu et al. (2015), Che et al. (2015), and Albers et al. (2014)
Risk Prediction/Biomarker Discovery	
Descriptive Atemporal	Gotz et al. (2011), Vellanki et al. (2014), and Miotto et al. (2016)
Descriptive Temporal	
Cross-Sectional	Reddy and Aggarwal (2015), Paxton et al. (2013a), Wilcox and Hripcsak (2003), Sarkar et al. (2012), Letham et al. (2013), Ebadollahi et al. (2010), Feldman and Chawla (2014), Stiglic et al. (2015), Ngufor et al. (2015), Byrd et al. (2014), and Liang et al. (2014)
Time-Agnostic Case Control or Cohort	Lin and Haug (2008), Yadav et al. (2016b), Zolfaghar et al. (2013), Caruana et al. (2015), Westra et al. (2011), Skevofilakas et al. (2010), Oztekin et al. (2009), Ghalwash and Obradovic (2014), Somanchi et al. (2015), Li et al. (2016a, 2016b, 2016c), Cheng et al. (2016), Collins et al. (2011), Simon et al. (2011a), Mani et al. (2014), Matheny et al. (2010), Pakhomov et al. (2011), Fung et al. (2008), Wells et al. (2008), Vinzamuri et al. (2014), Li et al. (2015), van der Heijden et al. (2014), Jackson et al. (2011), Paxton et al. (2013b), Zhao et al. (2011), Maroco et al. (2011), Breault et al. (2002), Wang et al. (2015b), Sun et al. (2014), and Dey et al. (2015)

(Continued)

Table 2. Continued

Time-Aware Case Control or Cohort	Xing et al. (2014), Wang et al. (2014b), Yu et al. (2008), Ghassempour et al. (2014), Yadav et al. (2015b), Gatti et al. (2011), Sandri et al. (2014), Vinzamuri and Reddy (2013), Peelen et al. (2010), Choi et al. (2015), Wang et al. (2013a), Luo et al. (2016), Hripcsak et al. (2015), and Hagar et al. (2014)
Quantifying the Effect of Intervention	
Descriptive Atemporal	
Descriptive Temporal	
Cross-Sectional	
Time-Agnostic Case Control or Cohort	Schrom et al. (2013), Westra et al. (2017), Yadav et al. (2016a), and Yadav (2017)
Time-Aware Case Control or Cohort	
Constructing Evidence-Based Guidelines	
Descriptive Atemporal	
Descriptive Temporal	Pivovarov et al. (2014)
Cross-Sectional	
Time-Agnostic Case Control or Cohort	Pruinelli et al. (2014), Mani et al (2007), and Pruinelli et al. (2015, 2016)
Time-Aware Case Control or Cohort	
Adverse Event Detection	
Descriptive Atemporal	
Descriptive Temporal	
Cross-Sectional	Sathyanarayana et al. (2014), White et al. (2013), Iyer et al. (2014), Haerian et al. (2012), Pathak et al. (2013), Carroll et al. (2011), Xu et al. (2011), Bobo et al. (2014), Zhang et al. (2015), and Melton and Hripcsak (2005)
Time-Agnostic Case Control or Cohort	
Time-Aware Case Control or Cohort	Hauskrecht et al. (2013)

we are interested in the reduction of mortality as a result of starting patients on a first-line drug *earlier*, then our control group should consist of patients who are not taking any drugs for the disease; if we are interested in the reduction of side effects by using a particular drug over the standard treatment, then the control group should consist of patients who take an equally effective alternative drug. Propensity score matching alone does *not* prevent us from selecting patients from the wrong control group, because comparable patients may exist in all of these groups.

Another study design consideration is length of exposure. For many interventions, relatively long exposure periods are needed to achieve the desired effect. Requiring a certain exposure period (e.g., we only consider patients with 180 days of exposure) introduces bias (known as immortality bias), because patients who did not tolerate the drug or died within 180 days are not included, potentially leading to an overly optimistic estimate for the effect of the intervention.

The second reason for the unpopularity of certain applications is their need for extensive collaborations. For example, constructing (or even just evaluating) evidence-based guidelines requires

more than just a study design and causal inference. It is decidedly interdisciplinary, often critically depending on interpretation and feedback from researchers with clinical expertise. To give a concrete example, consider the National Comprehensive Cancer Network guidelines for metastatic colon cancer: "Evidence increasingly suggests that [. . .] mutation makes response to [. . .], as single agents or in combination with cytotoxic chemotherapy, highly unlikely." This guideline neither requires nor prohibits the use of certain agents when a specific mutation is present, but merely draws attention to the possibility that the agent may not work. Such guidelines leave a lot of room for interpretation: do we administer the referenced drug or select a different one? Another example is the Surviving Sepsis Campaign 3-hour bundle. This bundle gives a list of actions, such as "administer broad spectrum antibiotics," that need to be carried out within 3 hours after the suspicion of sepsis. This guideline is prescriptive (we know exactly what needs to be done and when), but even this guideline leaves room for interpretation through the "suspicion of sepsis" phrase. We need to estimate when a well-trained clinical would "suspect" sepsis.

The third reason that some applications are unpopular is that they are unnatural. Examples include the usage of unsupervised data mining techniques for predicting complications, quantifying the effect of interventions, and adverse event detection. For such applications, patients are already grouped into cases and controls and therefore case-control, retrospective, or cross-sectional study designs would be more natural.

9.2 Study Designs

Substantially more work has been carried out in retrospective or case-control settings as compared to the descriptive setting. This stems from the nature of research in the medical domain, as research in medical sciences has hitherto been driven by predefined outcomes. The defining difference between case-control and descriptive designs is the existence of an outcome.

The outcome provides a focus to the case/control studies that does not exist in descriptive studies. The lack of an outcome combined with the high dimensionality and the associated heterogeneity of EHR data often lead to increased complexity, which translates to an exponentially large number of patterns. If the researchers tune their algorithm to find a small number of patterns, they tend to be trivial patterns; if they extract a large number of patterns, those patterns are often redundant and difficult to interpret. Developing algorithms to directly discover novel patterns is an area for future method development.

A well-chosen study design with appropriate comparison groups can help address this problem. Consider, for example, the difference between the retrospective case-control and cohort study. For the case-control design, we select cases (patients with the outcome in question) and controls (patients without the outcome in question) at the end of the study and follow them backward in time, examining their exposures. If the outcome is rare, this approach helps focus on a relatively small population. Exposures associated with the outcome are likely to be present in this smaller population in sufficient amounts, while "random" (unassociated) outcomes are reduced. With reduced "random" exposures, we can substantially reduce the number of uninteresting patterns. Conversely, if the exposure is rare (relative to the outcome), we would select a cohort study, where patients are selected based on their exposures at baseline and followed forward in time to see whether they develop the outcome in question. The rareness of the exposure will control the number of spurious patterns.

9.3 Methodologies

Little research has been done to utilize the temporal nature of EHR data. For example, descriptive atemporal studies are more frequent than descriptive temporal techniques. The situation is similar for retrospective and case-control studies. We identify a couple of reasons for this phenomenon.

First, the duration of EHR data available with healthcare providers rarely exceeds a couple of years. Diseases such as T2DM take around 5 to 10 years for patients to progress from one state to a state of advanced complication. With such a limited duration of data available, this progression cannot be studied effectively. Second, censoring and irregular EHR data limit the application of techniques to EHR data, since addressing these issues often requires sophisticated techniques, such as those from Section 7.2, and rigorous study designs.

The prevalent method of representing irregular time-series data is to add multiple observations per patient to the observation matrix. Doing so violates the i.i.d. (independent and identically distributed) assumption of observations, making them correlated. Section 7.2 describes a number of techniques for addressing this correlation, but a careful look at these techniques reveals that they are all regression models. The most popular data mining techniques (SVM, random forests, neural networks) are not capable of handling this situation satisfactorily without modification. Simply ignoring the correlation could result in reasonable estimates if the numbers of observations contributed by each patient are similar. In healthcare, however, this is rarely the case since sick patients contribute much more data (more frequent visits, more tests per visit) than relatively healthy patients. Thus, ignoring correlation among observations will lead to biased estimates (a form of sampling bias).

To make matters worse, many of these techniques use validation (e.g., cross-validation) to avoid or reduce overfitting (e.g., performance on a cross-validation set often serves as the stopping criterion for tree induction). The sampling unit for these techniques is typically each row of the observation matrix. However, when each patient can contribute multiple rows, the sampling unit has to be the patient (not the observation). Therefore, all observations of the same patient must either be in the training set or the validation set. We cannot have some observations in the training set and others in the validation set. Failing to follow this rule can lead to overfitting.

9.4 Barriers to Clinical Data Mining

This survey is a testament to the effort that the data mining community has expended on mining EHR data, yet the translation of these results into clinical practice is lagging. This lag is a direct consequence of what the authors of this survey view as the key difference between clinical data mining and data mining in general: clinical data mining models have to be validated to a standard that is much higher than in many other fields. Many fields, recommendation systems for example, "only" require that we demonstrate a reproducible performance advantage of the proposed model over the models in use. This standard of validation can be satisfied relatively easily through textbook validation techniques, such as cross-validation. Validity in healthcare often presumes that the model captures some knowledge of human physiology and pathophysiology. Without using predictors rooted in (patho)physiology for clinical prediction, we run the risk of capturing a factor that may change at any time, invalidating a model that appears to produce computationally reproducible results. A prototypical example of this phenomenon is the Google Flu Trend, which used queries related to influenza originating from a particular geographic region to assess the level of exposure to influenza in that region. As a result of changes in user query behavior, the model became invalid and this service is no longer offered.

With recent advances in machine learning (e.g., deep learning), understanding a model's decision process is becoming increasingly difficult. As the recent Workshop on Human Interpretability of Machine Learning (ICML 2016) shows, healthcare is not the only field that desires an understanding of the decision process of the machine-learned models. A paper by Lipton (2016) draws a parallel between model interpretability and trustworthiness and describes several criteria for a trustworthy model. Constructing trustworthy models is one direction that can help remove barriers from the implementation of machine-learned models in clinical practice.

10 CONCLUDING REMARKS

The current landscape of healthcare and the drivers that shape it virtually ensure that mining EHR data will play an increasingly important role in the future. Major examples of these drivers include the transition from the current reimbursement-based healthcare model to the Accountable Care Organization model; personalizing care to make it safer and more efficient and to reduce waste; and shared decision making, which reflects patients' desire to become more involved in their own care. To be successful, all of these drivers require strong analytics based on large populations.

The cornerstone of modern medicine is clinical evidence, and generating hypotheses for clinical evidence is one role that data mining is likely to play. This role is not a replacement of clinical trials, but rather a synergistic role, where data mining can create high-quality hypotheses that can be validated by conducting RCTs. Clinical trials are expensive; thus, the number of patients participating in a study is kept to the minimum required to validate the hypothesis. Secondary use of such small datasets for hypothesis generation is impractical, creating an opportunity for mining EHR data.

Even before the broad availability of EHR-based clinical data, large-scale observational studies over entire populations have been carried out. However, these studies have typically utilized claims data, which exhibit many of the key characteristics of EHR data, including the intermittent generation of longitudinal data, censoring, confounding, and the need for a robust study design. Methods utilized for these studies came from biostatistics and epidemiology. The integration of biostatistics techniques into clinical data mining (Cox regression along with its penalized variants) has already started, and novel extensions to biostatistics techniques using data mining are being proposed. A similar interchange of ideas with epidemiology needs to happen but is in an early stage.

To reach its full potential, EHR data mining needs to develop these methods further in many directions. One of these directions is data representation. EHR data is rich; it consists of highly heterogeneous data, collected from a wide range of sources (structured and unstructured clinical data, images, omics data, wearable sensors, mobile health), often at varying frequencies and resolutions. The emergence of such data-type heterogeneity necessitates that we revisit even fundamental questions, such as how data is best represented for modeling purposes. Other directions include analyzing temporal and sequence data, handling missing data, and causal inference. Both data mining and biostatistics/epidemiology have methods to address these issues, but they need to be further developed to suit EHR data better.

The characteristics of EHR data that drive the development of new data mining techniques are not unique to EHR data. Heterogeneity is present and poses challenges in many areas, including earth science and climate; time-to-event data finds its origins in failure analysis; and censoring along with the intermittent interactions with customers also happens in customer relationship management and recommendation systems. Mirroring how techniques for censored data and causal inference are being adopted from biostatistics and study designs from epidemiology into clinical data mining, new developments in EHR mining will likely find applications in many other data mining and analytics domains.

Development of interpretable and intelligible models is another interesting research area. Caruana et al. (2015) developed a novel high-performance generalized additive model with pairwise interactions for the pneumonia risk prediction problem. They claimed that their method has the same accuracy as neural networks and random forests while being simultaneously as interpretable as logistic regression and Naive Bayes models. By achieving state-of-the-art accuracy combined with the ability to extract information from complex EHRs, their technique provides a platform to extract meaningful information and apply it in practice.

REFERENCES

J. Abramson et al. 2001. *Making Sense of Data: A Self-Instruction Manual on the Interpretation of Epidemiological Data.* Oxford University Press.

R. Agrawal et al. 1994. Fast algorithms for mining association rules. In *Proc. 20th Int. Conf. Very Large Data Dases, VLDB*, Vol. 1215. 487–499.

D. Albers and G. Hripcsak. 2010. A statistical dynamics approach to the study of human health data: Resolving population scale diurnal variation in laboratory data. *Physics Letters A* 374, 9 (2010), 1159–1164.

D. Albers and G. Hripcsak. 2012. Using time-delayed mutual information to discover and interpret temporal correlation structure in complex populations. *Chaos* 22, 1 (2012), 013111.

D. Albers et al. 2014. Dynamical phenotyping: Using temporal analysis of clinically collected physiologic data to stratify populations. *PloS One* 9, 6 (2014), e96443.

C. Aliferis et al. 2010. Local causal and Markov blanket induction for causal discovery and feature selection for classification part i: Algorithms and empirical evaluation. *Journal of Machine Learning Research* 11, (Jan. 2010), 171–234.

D. Anastasiu et al. 2016. Big data and recommender systems.

P. Austin et al. 2012. Regression trees for predicting mortality in patients with cardiovascular disease: What improvement is achieved by using ensemble-based methods? *Biometrical Journal* 54, 5 (Sept. 2012), 657–673.

L. Babuin et al. 2008. Elevated cardiac troponin is an independent risk factor for short-and long-term mortality in medical intensive care unit patients. *Critical Care Medicine* 36, 3 (2008), 759–765.

S. Barretto et al. 2003. Linking guidelines to electronic health record design for improved chronic disease management. In *Proc. of AMIA*, Vol. 2003. 66.

I. Batal and M. Hauskrecht. 2010. Mining clinical data using minimal predictive rules. *AMIA Annual Symposium Proceedings*, Vol. 2010, 31–35.

I. Batal, L. Sacchi, R. Bellazzi, and M. Hauskrecht. 2009. Multivariate time series classification with temporal abstractions. University of Pittsburgh.

I. Batal et al. 2012. Mining recent temporal patterns for event detection in multivariate time series data. *Proc. of KDD* (2012), 280.

A. Bauer-Mehren et al. 2013. Network analysis of unstructured EHR data for clinical research. *AMIA Summit on Translational Science* 2013 (2013), 14–8.

A. Bitton and T. Gaziano. 2010. The Framingham heart study's impact on global risk assessment. *Progress in Cardiovascular Diseases* 53, 1 (2010), 68–78.

W. Bobo et al. 2014. An electronic health record driven algorithm to identify incident antidepressant medication users. *Journal of the American Medical Informatics Association* 21, 5 (2014), 785–791.

J. L. Breault, C. R. Goodall, and P. J. Fos. 2002. Data mining a diabetic data warehouse. *AIM* 26, 1 (2002), 37–54.

R. Byrd et al. 2014. Automatic identification of heart failure diagnostic criteria, using text analysis of clinical notes from electronic health records. *IJMI* 83, 12 (2014), 983–992.

H. Cao et al. 2005. Mining a clinical data warehouse to discover disease-finding associations using co-occurrence statistics. *AMIA* 2005 (Jan. 2005), 106–110.

R. Carroll et al. 2011. Naïve electronic health record phenotype identification for rheumatoid arthritis. *AMIA* 2011 (Jan. 2011), 189–196.

R. Caruana et al. 2015. Intelligible models for healthcare: Predicting pneumonia risk and hospital 30-day readmission. In *Proc. of KDD*. ACM, 1721–1730.

M. R. Castro et al. 2016. Statin use, diabetes incidence and overall mortality in normoglycemic and impaired fasting glucose patients. *Journal of General Internal Medicine* 31, 5 (2016), 502–508.

Y. Chang et al. 2011. Predicting hospital-acquired infections by scoring system with simple parameters. *PloS One* 6, 8 (Jan. 2011), e23137.

Z. Che et al. 2015. Deep computational phenotyping. In *Proc. of KDD*. 507–516.

Z. Che et al. 2017. Exploiting convolutional neural network for risk prediction with medical feature embedding. *arXiv preprint arXiv:1701.07474* (2017).

R. Chen et al. 2016. Patient stratification using electronic health records from a chronic disease management program. *IEEE Journal of Biomedical and Health Informatics*. IEEE.

Y. Chen et al. 2015. Building bridges across electronic health record systems through inferred phenotypic topics. *JBI* 55 (2015), 82–93.

Y. Cheng, F. Wang, P. Zhang, and J. Hu. 2016. Risk prediction with electronic health records: A deep learning approach. In *Proceedings of the 2016 SIAM International Conference on Data Mining*. SIAM, 432–440.

Y. Cheng et al. 2016. Risk prediction with electronic health records: A deep learning approach. In *Proc. of SIAM-SDM*. 432–440.

E. Choi, M. T. Bahadori, and J. Sun. 2015. Doctor AI: Predicting clinical events via recurrent neural networks. *arXiv preprint arXiv:1511.05942* (2015).
E. Choi et al. 2015. Doctor AI: Predicting clinical events via recurrent neural networks. *arXiv preprint arXiv:1511.05942* (2015).
E. Choi et al. 2016. Multi-layer representation learning for medical concepts. *arXiv preprint arXiv:1602.05568* (2016).
S. Cholleti et al. 2012. Leveraging derived data elements in data analytic models for understanding and predicting hospital readmissions. *AMIA* 2012 (Jan. 2012), 103–111.
C. Chung et al. 1991. Survival analysis: A survey. *Journal of Quantitative Criminology* 7, 1 (1991), 59–98.
M. Coleman et al. 2008. CONCORD. *Lancet Oncology* 9, 8 (Aug. 2008), 730–756.
G. Collins et al. 2011. Developing risk prediction models for type 2 diabetes: A systematic review of methodology and reporting. *BMC Medicine* 9, 1 (Jan. 2011), 103.
D. R Cox. 1992. Regression models and life-tables. In *Breakthroughs in Statistics*. Springer, 527–541.
D. Dasgupta and N. Chawla. 2014. Disease and medication networks: An insight into disease-drug interactions. In *2nd International Conference on Big Data and Analytics in Healthcare*. Singapore.
R. Dehejia and S. Wahba. 2002. Propensity score-matching methods for nonexperimental causal studies. *Review of Economics and Statistics* 84, 1 (2002), 151–161.
D. Demner-Fushman et al. 2009. What can natural language processing do for clinical decision support? *JBI* 42, 5 (2009), 760–772.
S. Dey et al. 2015. Predicting the factors of improvement of health status of home health care patients: A holistic data mining approach. In *AMIA*.
J. Dormandy et al. 2005. Secondary prevention of macrovascular events in patients with type 2 diabetes in the proactive study (prospective pioglitazone clinical trial in macrovascular events): A randomised controlled trial. *Lancet* 366, 9493 (2005), 1279–1289.
F. Doshi-Velez et al. 2014. Comorbidity clusters in autism spectrum disorders: An electronic health record time-series analysis. *Pediatrics* 133, 1 (Jan. 2014), e54–63.
G. Dunteman. 1989. *Principal Components Analysis*. Number 69. Sage.
S. Ebadollahi et al. 2010. Predicting patients trajectory of physiological data using temporal trends in similar patients: A system for near-term prognostics. In *Proc. of AMIA*, Vol. 2010. AMIA, 192.
E. Ekinci et al. 2011. Dietary salt intake and mortality in patients with type 2 diabetes. *Diabetes Care* 34, 3 (2011), 703–709.
R. Epstein et al. 2013. Automated identification of drug and food allergies entered using non-standard terminology. *JAMIA* 20, 5 (2013), 962–968.
L. Evers and C. Messow. 2008. Sparse kernel methods for high-dimensional survival data. *Bioinformatics* 24, 14 (2008), 1632–1638.
K. Feldman and N. Chawla. 2014. Admission duration model for infant treatment (ADMIT). In *Proc. of BIBM*. IEEE, 583–587.
M. Field et al. 1992. *Guidelines for Clinical Practice: From Development to Use*. National Academies Press.
E. Funai et al. 2001. Distribution of study designs in four major US journals of obstetrics and gynecology. *Gynecologic and Obstetric Investigation* 51, 1 (2001), 8–11.
G. Fung et al. 2008. Privacy-preserving predictive models for lung cancer survival analysis. *Practical Privacy-Preserving Data Mining* (2008), 40.
E. Gatti et al. 2011. A continuous time Bayesian network model for cardiogenic heart failure. *Flexible Services and Manufacturing Journal* 24, 4 (Dec. 2011), 496–515.
M. Ghalwash and Z. Obradovic. 2014. A data-driven model for optimizing therapy duration for septic patients. In *Proc. of SDM-DMHH*.
S. Ghassempour et al. 2014. Clustering multivariate time series using hidden Markov models. *International Journal of Environmental Research and Public Health* 11, 3 (2014), 2741–2763.
L. Gordon and R. A. Olshen. 1985. Tree-structured survival analysis. *Cancer Treatment Reports* 69, 10 (1985), 1065–1069.
D. Gotz. 2016. Adaptive contextualization: Combating bias during high-dimensional visualization and data selection. In *Proc. of ICIUI*. ACM, 85–95.
D. Gotz and H. Stavropoulos. 2014. Decisionflow: Visual analytics for high-dimensional temporal event sequence data. *TVGC* 20, 12 (2014), 1783–1792.
D. Gotz et al. 2011. Visual cluster analysis in support of clinical decision intelligence. *AMIA* 2011 (Jan. 2011), 481–490.
D. Gotz et al. 2014. A methodology for interactive mining and visual analysis of clinical event patterns using electronic health record data. *JBI* 48 (2014), 148–159.
D. Goyal et al. 2012. Clinically identified postpartum depression in asian american mothers. *JOGNN/NAACOG* 41, 3 (2012), 408–416.
D. Grimes and K. Schulz. 2002. An overview of clinical research: The lay of the land. *Lancet* 359, 9300 (2002), 57–61.

S. Gruber et al. 1986. Clinical epidemiology: The architecture of clinical research. *Yale Journal of Biology and Medicine* 59, 1 (1986), 77.
K. Haerian et al. 2012. Detection of pharmacovigilance-related adverse events using electronic health records and automated methods. *Clinical Pharmacology and Therapeutics* 92, 2 (Aug. 2012), 228–234.
Y. Hagar et al. 2014. Survival analysis with electronic health record data: Experiments with chronic kidney disease. *Statistical Analysis and Data Mining* 7, 5 (2014), 385–403.
D. Hanauer and N. Ramakrishnan. 2013. Modeling temporal relationships in large scale clinical associations. *JAMIA* 20, 2 (2013), 332–341.
D. Hanauer et al. 2009. Exploring clinical associations using '-omics' based enrichment analyses. *PloS One* 4, 4 (Jan. 2009), e5203.
D. Hanauer et al. 2013. Describing the relationship between cat bites and human depression using data from an electronic health record. *PLoS One* 8, 8 (2013), e70585.
M. Hauskrecht et al. 2013. Outlier detection for patient monitoring and alerting. *JBI* 46, 1 (2013), 47–55.
M. Hearst et al. 1998. Support vector machines. *ISA* 13, 4 (1998), 18–28.
C. Hennekens et al. 1987. *Epidemiology in Medicine*. Boston: Little Brown and Company, 1987.
V. Herasevich et al. 2013. Connecting the dots: Rule-based decision support systems in the modern EMR era. *Journal of Clinical Monitoring and Computing* 27, 4 (Aug. 2013), 443–448.
J. Hogan and T. Lancaster. 2004. Instrumental variables and inverse probability weighting for causal inference from longitudinal observational studies. *SMMR* 13, 1 (2004), 17–48.
A. Holmes et al. 2011. Discovering disease associations by integrating electronic clinical data and medical literature. *PloS One* 6, 6 (Jan. 2011), e21132.
G. Hripcsak. 2013. Correlating electronic health record concepts with healthcare process events. *JAMIA* 20, e2 (Dec. 2013), e311–318.
G. Hripcsak and D. Albers. 2013. Next-generation phenotyping of electronic health records. *JAMIA* 20, 1 (2013), 117–121.
G. Hripcsak et al. 1997. Automated tuberculosis detection. *JAMIA* 4, 5 (1997), 376–381.
G. Hripcsak et al. 2011. Bias associated with mining electronic health records. *Journal of Biomedical Discovery and Collaboration* 6 (2011), 48.
G. Hripcsak et al. 2015. Parameterizing time in electronic health record studies. *Journal of the American Medical Informatics Association* 22, 4 (2015), 794–804.
S. Huang et al. 2014. Toward personalizing treatment for depression: Predicting diagnosis and severity. *Journal of the American Medical Informatics Association* 21, 6 (2014), 1069–1075.
K. Ikeda et al. 1991. Effect of repeated transcatheter arterial embolization on the survival time in patients with hepatocellular carcinoma. An analysis by the Cox proportional hazard model. *Cancer* 68, 10 (1991), 2150–2154.
H. Ishwaran et al. 2008. Random survival forests. *The Annals of Applied Statistics* (2008), 841–860.
S. Iyer et al. 2014. Mining clinical text for signals of adverse drug-drug interactions. *JAMIA* 21, 2 (2014), 353–62.
H. Jackson et al. 2011. Data mining derived treatment algorithms from the electronic medical record improve theoretical empirical therapy for outpatient urinary tract infections. *Journal of Urology* 186, 6 (2011), 2257–2262.
H. Jin et al. 2008. Mining unexpected temporal associations: Applications in detecting adverse drug reactions. *TITB* 12, 4 (July 2008), 488–500.
L. Kalankesh et al. 2013. Taming EHR data: Using semantic similarity to reduce dimensionality. *Studies in Health Technology and Informatics* 192 (Jan. 2013), 52–56.
E. Kaplan and P. Meier. 1958. Nonparametric estimation from incomplete observations. *JASA* 53, 282 (1958), 457–481.
S. Karnik et al. 2012. Predicting atrial fibrillation and flutter using electronic health records. *EMBC* 2012 (Jan. 2012), 5562–5565.
M. Kattan et al. 1998. Experiments to determine whether recursive partitioning (CART) or an artificial neural network overcomes theoretical limitations of Cox proportional hazards regression. *Computers and Biomedical Research* 31, 5 (1998), 363–373.
E. Kawaler et al. 2012. Learning to predict post-hospitalization VTE risk from EHR data. *AMIA* 2012 (Jan. 2012), 436–445.
N. Keiding et al. 1997. The role of frailty models and accelerated failure time models in describing heterogeneity due to omitted covariates. *SIM* 16, 2 (1997), 215–224.
J. Kelsey. 1996. *Methods in Observational Epidemiology*. Vol. 26. Oxford University Press.
F. Khan and V. Zubek. 2008. Support vector regression for censored data (SVRc): A novel tool for survival analysis. In *Proc. of ICDM*. IEEE, 863–868.
A. Khosla et al. 2010. An integrated machine learning approach to stroke prediction. In *Proc. of KDD*. ACM, 183–192.
J. Kirby et al. 2016. PheKB: A catalog and workflow for creating electronic phenotype algorithms for transportability. *Journal of the American Medical Informatics Association* 23, 6 (2016), 1046–1052.

J. Klein and M. Moeschberger. 2005. *Survival Analysis: Techniques for Censored and Truncated Data*. Springer Science & Business Media.
N. Laird and J. Ware. 1982. Random-effects models for longitudinal data. *Biometrics* (1982), 963–974.
T. Lasko et al. 2013. Computational phenotype discovery using unsupervised feature learning over noisy, sparse, and irregular clinical data. *PloS One* 8, 6 (2013), e66341.
J. Last et al. 2001. *A Dictionary of Epidemiology*. Vol. 141. Oxford University Press.
J. Lee. 1994. Odds ratio or relative risk for cross-sectional data? *International Journal of Epidemiology* 23, 1 (1994), 201–203.
B. Letham et al. 2013. An interpretable stroke prediction model using rules and Bayesian analysis. *AAAI (Late-Breaking Developments)*.
D. Levinson and Inspector General. 2010. Adverse events in hospitals: National incidence among medicare beneficiaries. *Department of Health and Human Services Office of the Inspector General* (2010).
K. Levit et al. 2003. Trends in US health care spending, 2001. *Health Affairs* 22, 1 (2003), 154–164.
Y. Li et al. 2014. A method for controlling complex confounding effects in the detection of adverse drug reactions using electronic health records. *JAMIA* 21, 2 (2014), 308–314.
Y. Li et al. 2015. Constrained elastic net based knowledge transfer for healthcare information exchange. *DMKD* 29, 4 (2015), 1094–1112.
Y. Li et al. 2016a. A distributed ensemble approach for mining healthcare data under privacy constraints. *Information Sciences* 330 (2016), 245–259.
Y. Li et al. 2016b. Regularized parametric regression for high-dimensional survival analysis. In *Proceedings of SIAM International Conference on Data Mining*.
Y. Li et al. 2016c. Regularized weighted linear regression for high-dimensional censored data. In *Proceedings of SIAM International Conference on Data Mining*. SIAM.
K. Liang et al. 1990. The Cox proportional hazards model with change point: An epidemiologic application. *Biometrics* (1990), 783–793.
Z. Liang et al. 2014. Deep learning for healthcare decision making with EMRs. In *Proc. of BIBM*. 556–559.
V. Liao and M. Chen. 2013. Efficient mining gapped sequential patterns for motifs in biological sequences. *BMC Systems Biology* 7, Suppl. 4 (Jan. 2013), S7.
D. Lilienfeld and P. Stolley. 1994. *Foundations of Epidemiology*. Oxford University Press.
J. Lin and P. Haug. 2008. Exploiting missing clinical data in Bayesian network modeling for predicting medical problems. *JBI* 41, 1 (2008), 1–14.
Z. Lipton. 2016. The mythos of model interpretability. *arXiv preprint arXiv:1606.03490* (2016).
C. Liu et al. 2015. Temporal phenotyping from longitudinal electronic health records: A graph based framework. In *Proc. of KDD*. 705–714.
P. Lucas et al. 2004. Bayesian networks in biomedicine and health-care. *AIM* 30, 3 (March 2004), 201–214.
T. Lumley et al. 2002. A stroke prediction score in the elderly: Validation and web-based application. *Journal of Clinical Epidemiology* 55, 2 (2002), 129–136.
Y. Luo et al. 2016. Tensor factorization toward precision medicine. *Briefings in Bioinformatics* 18, 3 (2016), 511–514.
S. Mani et al. 2007. Learning causal and predictive clinical practice guidelines from data. In *Proceedings of the 12th World Congress on Health (Medical) Informatics; Building Sustainable Health Systems (Medinfo'07)*. IOS Press, 850.
S. Mani et al. 2012. Type 2 diabetes risk forecasting from EMR data using machine learning. *AMIA* 2012 (Jan. 2012), 606–615.
S. Mani et al. 2014. Medical decision support using machine learning for early detection of late-onset neonatal sepsis. *JAMIA* 21, 2 (2014), 326–336.
C. J. Mann. 2003. Observational research methods. Research design II: cohort, cross sectional, and case-control studies. *Emergency Medicine Journal* 20, 1 (2003), 54–60.
B. Marlin et al. 2012. Unsupervised pattern discovery in electronic health care data using probabilistic clustering models. In *Proc. of SIGHIT*. ACM, 389–398.
J. Maroco et al. 2011. Data mining methods in the prediction of dementia: A real-data comparison of the accuracy, sensitivity and specificity of linear discriminant analysis, logistic regression, neural networks, support vector machines, classification trees and random forests. *BMC Research Notes* 4 (Jan. 2011), 299.
M. Matheny et al. 2010. Development of inpatient risk stratification models of acute kidney injury for use in electronic health records. *Medical Decision Making* 30, 6 (2010), 639–650.
J. Matthews. 2006. *Introduction to Randomized CCT*. CRC Press.
I. Melnyk et al. 2013. Detection of precursors to aviation safety incidents due to human factors. In *2013 IEEE 13th International Conference on Data Mining Workshops (ICDMW'13)*. IEEE, 407–412.
G. Melton and G. Hripcsak. 2005. Automated detection of adverse events using natural language processing of discharge summaries. *JAMIA* 12, 4 (2005), 448–457.
M. Mercer et al. 2009. Multimorbidity in primary care: Developing the research agenda. *Family Practice* 26, 2 (2009), 79–80.

R. Miotto et al. 2016. Deep patient: An unsupervised representation to predict the future of patients from the electronic health records. *Scientific Reports* 6 (2016).
D. Moher et al. 1998. Does quality of reports of randomised trials affect estimates of intervention efficacy reported in meta-analyses? *Lancet* 352, 9128 (1998), 609–613.
K. Monsen et al. 2010. Discovering client and intervention patterns in home visiting data. *WJNR* 32, 8 (2010), 1031–1054.
K. Monsen et al. 2015. Factors explaining variability in health literacy outcomes of public health nursing clients. *Public Health Nursing* 32, 2 (2015), 94–100.
M. Munson et al. 2014. Data mining for identifying novel associations and temporal relationships with charcot foot. *Journal of Diabetes Research* 2014 (2014).
S. Nachimuthu et al. 2010. Modeling glucose homeostasis and insulin dosing in an intensive care unit using dynamic Bayesian networks. *AMIA* 2010 (Jan. 2010), 532–536.
G. Nadkarni et al. 2014. Development and validation of an electronic phenotyping algorithm for chronic kidney disease. In *Proc. of AMIA*, Vol. 2014. 907.
K. Ng et al. 2014. PARAMO: A parallel predictive modeling platform for healthcare analytic research using electronic health records. *JBI* 48 (2014), 160–170.
C. Ngufor et al. 2015. A heterogeneous multi-task learning for predicting RBC transfusion and perioperative outcomes. In *Proc. of AIM*. Springer, 287–297.
E. Nomura et al. 2003. Population-based study of relationship between hospital surgical volume and 5-year survival of stomach cancer patients in Osaka, Japan. *Cancer Science* 94, 11 (Nov. 2003), 998–1002.
W. Oh et al. 2016. Type 2 diabetes mellitus trajectories and associated risk factors. *Big Data* 4, 1 (2016), 25–30.
H. Ory. 1977. Association between oral contraceptives and myocardial infarction: A review. *JAMA* 237, 24 (1977), 2619–2622.
A. Oztekin et al. 2009. Predicting the graft survival for heart-lung transplantation patients: An integrated data mining methodology. *IJMI* 78, 12 (Dec. 2009), e84–96.
S. Pakhomov et al. 2011. The role of the electronic medical record in the assessment of health related quality of life. *AMIA* 2011 (Jan. 2011), 1080–1088.
M. Panahiazar et al. 2015. Using EHRs and machine learning for heart failure survival analysis. In *MEDINFO*, Vol. 216. IOS Press, 40.
Y. Park and J. Ghosh. 2014. A hierarchical ensemble of α-trees for predicting expensive hospital visits. In *Brain Informatics and Health*. Springer, 178–187.
J. Pathak et al. 2012. Applying semantic web technologies for phenome-wide scan using an electronic health record linked biobank. *JBS* 3 (2012), 10.
J. Pathak et al. 2013. Electronic health records-driven phenotyping: Challenges, recent advances, and perspectives. *JAMIA* 20, e2 (Dec. 2013), e206–211.
D. Patnaik et al. 2011. Experiences with mining temporal event sequences from electronic medical records: Initial successes and some challenges. In *Proc. of KDD*. 360–368.
C. Paxton et al. 2013a. Developing predictive models using electronic medical records: Challenges and pitfalls. In *AMIA*, Vol. 2013. 1109.
C. Paxton et al. 2013b. Developing predictive models using electronic medical records: Challenges and pitfalls. *AMIA* 2013 (Jan. 2013), 1109–1115.
L. Peelen et al. 2010. Using hierarchical dynamic Bayesian networks to investigate dynamics of organ failure in patients in the intensive care unit. *JBI* 43, 2 (Apr. 2010), 273–286.
D. Peikes et al. 2008. Propensity score matching: A note of caution for evaluators of social programs. *The American Statistician* 62, 3 (2008), 222–231.
P. Peissig et al. 2012. Importance of multi-modal approaches to effectively identify cataract cases from electronic health records. *JAMIA* 19, 2 (2012), 225–234.
A. Perotte and G. Hripcsak. 2013. Temporal properties of diagnosis code time series in aggregate. *JBHI* 17, 2 (2013), 477–483.
R. Pivovarov et al. 2014. Temporal trends of hemoglobin A1c testing. *JAMIA* 21, 6 (2014), 1038–1044.
K. Polkinghorne et al. 2004. Vascular access and all-cause mortality: A propensity score analysis. *JASN* 15, 2 (2004), 477–486.
L. Pruinelli et al. 2014. Data mining methodologies to discover best practices for diabetic patients with health disparities. In *AMIA*.
L. Pruinelli et al. 2015. Clustering health data to discover EBP interventions for sepsis prevention and treatment for health disparities. In *AMIA*.
L. Pruinelli et al. 2016. A data mining approach to determine sepsis guideline impact on inpatient mortality and complications. *AMIA Summits on Translational Science Proceedings* 2016 (2016), 194.
A. Ramirez et al. 2012. Predicting warfarin dosage in European-Americans and African-Americans using DNA samples linked to an electronic health record. *Pharmacogenomics* 13, 4 (March 2012), 407–418.

S. Rana et al. 2015. A predictive framework for modeling healthcare data with evolving clinical interventions. *Statistical Analysis and Data Mining* 8, 3 (2015), 162–182.
R. Ranganath et al. 2016. Deep survival analysis. In *Machine Learning for Healthcare Conference*. 101–114.
C. Reddy and C. Aggarwal. 2015. *Healthcare Data Analytics*. Vol. 36. CRC Press.
C. Rihal et al. 2002. Incidence and prognostic importance of acute renal failure after percutaneous coronary intervention. *Circulation* 105, 19 (2002), 2259–2264.
J. Robins et al. 2000. Marginal structural models and causal inference in epidemiology. LWW.
J. Robinson et al. 2011. Lack of association between 25 (OH) D levels and incident type 2 diabetes in older women. *Diabetes Care* 34, 3 (2011), 628–634.
J. Rodrigues. 2009. *Health Information Systems: Concepts, Methodologies, Tools, and Applications: Concepts, Methodologies, Tools, and Applications*. Vol. 1. IGI Global.
F. Roque et al. 2011. Using electronic patient records to discover disease correlations and stratify patient cohorts. *PLoS Computational Biology* 7, 8 (Aug. 2011), e1002141.
C. Rose et al. 2005. A dynamic Bayesian network for handling uncertainty in a decision support system adapted to the monitoring of patients treated by hemodialysis. In *17th IEEE International Conference on Tools with Artificial Intelligence (ICTAI'05)*. IEEE, 5–pp.
P. Rossing et al. 1996. Predictors of mortality in insulin dependent diabetes: 10 year observational follow up study. *BMJ* 313, 7060 (1996), 779–784.
K. Rothman and S. Greenland. 1986. Modern epidemiology.Boston: Little brown and company. *Study Design for Comparison of Orthoses* 273 (1986), 101–106.
L. Sacchi et al. 2007. Data mining with temporal abstractions: Learning rules from time series. *DMKD* 15, 2 (June 2007), 217–247.
D. Sackett. 2000. *Evidence-Based Medicine*. Wiley Online Library.
D. L Sackett. 1979. Bias in analytic research. *Journal of Chronic Diseases* 32, 1 (1979), 51–63.
M. Sandri et al. 2014. Dynamic Bayesian networks to predict sequences of organ failures in patients admitted to ICU. *JBI* 48 (Apr. 2014), 106–113.
C. Sarkar and J. Srivastava. 2013. Impact of density of lab data in EHR for prediction of potentially preventable events. In *IEEE ICHI*. 529–534.
C. Sarkar et al. 2012. Improved feature selection for hematopoietic cell transplantation outcome prediction using rank aggregation. In *Proc. of FedCSIS*. IEEE, 221–226.
A. Sarvestani et al. 2010. Predicting breast cancer survivability using data mining techniques. In *2010 2nd International Conference on Software Technology and Engineering*, Vol. 2. IEEE, V2–227–V2–231.
A. Sathyanarayana et al. 2014. Clinical decision making: A framework for predicting rx response. In *Proc. of ICDM-W*. IEEE, 1185–1188.
J. Schrom et al. 2013. Quantifying the effect of statin use in pre-diabetic phenotypes discovered through association rule mining. *AMIA* 2013 (Jan. 2013), 1249–1257.
P. Schulam et al. 2015. Clustering longitudinal clinical marker trajectories from electronic health data: Applications to phenotyping and endotype discovery. In *IEEE, Proc. of AAAI*.
M. Schuster et al. 1998. How good is the quality of health care in the united states? *Milbank Quarterly* 76, 4 (1998), 517–563.
D. Sengupta and P. Naik. 2013. SN algorithm: Analysis of temporal clinical data for mining periodic patterns and impending augury. *JCB* 3, 1 (Jan. 2013), 24.
Y. Shahar. 1997. A framework for knowledge-based temporal abstraction. *Artificial Intelligence* 90, 1–2 (Feb. 1997), 79–133.
M. Sharma et al. 2015. Feature-based factorized bilinear similarity model for cold-start top-n item recommendation. In *SIAM-SDM*. 190–198.
H. Shiao and V. Cherkassky. 2014. Learning using privileged information (LUPI) for modeling survival data. In *Proc. of IJCNN*. IEEE, 1042–1049.
B. Shickel et al. 2017. Deep EHR: A survey of recent advances on deep learning techniques for electronic health record (EHR) analysis. *arXiv Preprint arXiv:1706.03446* (2017).
A. Shin et al. 2010. Diagnostic analysis of patients with essential hypertension using association rule mining. *Healthcare Informatics Research* 16, 2 (June 2010), 77–81.
C. Shivade et al. 2014. A review of approaches to identifying patient phenotype cohorts using electronic health records. *JAMIA* 21, 2 (2014), 221–230.
P. Shivaswamy et al. 2007. A support vector approach to censored targets. In *Proc. of ICDM*. IEEE, 655–660.
G. Simon et al. 2011a. A simple statistical model and association rule filtering for classification. In *Proc. of KDD*. ACM Press, New York, 823.
G. Simon et al. 2013. Survival association rule mining towards type 2 diabetes risk assessment. *AMIA* 2013 (Jan. 2013), 1293–1302.

N. Simon et al. 2011b. Regularization paths for Coxs proportional hazards model via coordinate descent. *Journal of Statistical Software* 39, 5 (2011), 1–13.
M. Skevofilakas et al. 2010. A hybrid decision support system for the risk assessment of retinopathy development as a long term complication of type 1 diabetes mellitus. *EMBC* 2010 (Jan. 2010), 6713–6716.
P. Snow et al. 1994. Artificial neural networks in the diagnosis and prognosis of prostate cancer: A pilot study. *Journal of Urology* 152, 5 Pt 2 (1994), 1923–1926.
S. Somanchi et al. 2015. Early prediction of cardiac arrest (code blue) using electronic medical records. In *Proc. of KDD*. 2119–2126.
P. Stang et al. 2010. Advancing the science for active surveillance: Rationale and design for the observational medical outcomes partnership. *Annals of Internal Medicine* 153, 9 (2010), 600–606.
G. Stiglic et al. 2015. Comprehensible predictive modeling using regularized logistic regression and comorbidity based features. *PloS One* 10, 12 (2015), e0144439.
J. Sun et al. 2014. Predicting changes in hypertension control using electronic health records from a chronic disease management program. *JAMIA* 21, 2 (2014), 337–344.
Y. Sverchkov et al. 2012. A multivariate probabilistic method for comparing two clinical datasets. In *Proc. of SIGHIT*. ACM, 795–800.
P. Tan et al. 2006. *Introduction to Data Mining*. Vol. 1. Pearson Addison Wesley Boston.
T. Therneau and C. Crowson. 2014. Using time dependent covariates and time dependent coefficients in the Cox model. *Red* 2 (2014), 1.
R. Tibshirani et al. 1997. The lasso method for variable selection in the Cox model. *Statistics in Medicine* 16, 4 (1997), 385–395.
C. Torio et al. 2006. Trends in Potentially Preventable Hospital Admissions Among Adults and Children, 2005–2010: Statistical Brief# 151.
V. Van Bellev et al. 2007. Support vector machines for survival analysis. In *Proc. of CIMED*. 1–8.
M. van der Heijden et al. 2014. Learning Bayesian networks for clinical time series analysis. *JBI* 48 (Apr. 2014), 94–105.
P. Vellanki et al. 2014. Nonparametric discovery of learning patterns and autism subgroups from therapeutic data. In *ICPR*. IEEE, 1828–1833.
M. Verduijn et al. 2007. Temporal abstraction for feature extraction: A comparative case study in prediction from intensive care monitoring data. *AIM* 41, 1 (Sept. 2007), 1–12.
L. Vernick et al. 1984. Selection of neighborhood controls: Logistics and fieldwork. *Journal of Chronic Diseases* 37, 3 (1984), 177–182.
S. Vilar et al. 2012. Enhancing adverse drug event detection in electronic health records using molecular structure similarity: Application to pancreatitis. *PloS One* 7, 7 (Jan. 2012), e41471.
B. Vinzamuri and C. Reddy. 2013. Cox regression with correlation based regularization for electronic health records. In *Proc. of ICDM*. IEEE, 757–766.
B. Vinzamuri et al. 2014. Active learning based survival regression for censored data. In *Proc. of CIKM*. ACM, 241–250.
S. Wacholder et al. 1992. Selection of controls in case-control studies: II. Types of controls. *AJE* 135, 9 (1992), 1029–1041.
J. J. Walline. 2001. Designing Clinical Research: An Epidemiologic Approach. LWW.
F. Wang et al. 2013a. A framework for mining signatures from event sequences and its applications in healthcare data. *IEEE PAMI* 35, 2 (2013), 272–285.
F. Wang et al. 2014a. Clinical risk prediction with multilinear sparse logistic regression. In *SIGKDD*. ACM, 145–154.
X. Wang et al. 2013b. Exploring patient risk groups with incomplete knowledge. In *Proc. of ICDM*. IEEE, 1223–1228.
X. Wang et al. 2014b. Unsupervised learning of disease progression models. In *Proc. of KDD*. ACM, 85–94.
Y. Wang et al. 2015a. Rubik: Knowledge guided tensor factorization and completion for health data analytics. In *Proc. of KDD*. 1265–1274.
Z. Wang et al. 2015b. Dynamic poisson autoregression for influenza-like-illness case count prediction. In *Proc. of KDD*. ACM, 1285–1294.
B. Wells et al. 2008. Predicting 6-year mortality risk in patients with type 2 diabetes. *Diabetes Care* 31, 12 (2008), 2301–2306.
V. West et al. 2015. Innovative information visualization of electronic health record data: A systematic review. *JAMIA* 22, 2 (2015), 330–339.
B. Westra et al. 2011. Interpretable predictive models for knowledge discovery from home-care electronic health records. *JHE* 2, 1 (2011), 55–74.
B. Westra et al. 2017. Secondary analysis of an electronic surveillance system combined with multi-focal interventions for early detection of sepsis. *ACI* 8, 1 (2017), 47–66.
R. White et al. 2013. Web-scale pharmacovigilance: Listening to signals from the crowd. *JAMIA* 20, 3 (May 2013), 404–408.
A. Wilcox and G. Hripcsak. 2003. The role of domain knowledge in automating medical text report classification. *JAMIA* 10, 4 (2003), 330–338.

P. Wilson et al. 2008. Prediction of first events of coronary heart disease and stroke with consideration of adiposity. *Circulation* 118, 2 (2008), 124–130.
J. Wolff. 2002. Prevalence, expenditures and complications of multiple chronic conditions in the elderly. *Archives of Internal Medicine* 162, 20 (2002), 2269–2276.
J. Wooldridge. 1992. Some alternatives to the box-Cox regression model. *International Economic Review* (1992), 935–955.
A. Wright et al. 2010. An automated technique for identifying associations between medications, laboratory results and problems. *JBI* 43, 6 (Dec. 2010), 891–901.
Z. Xing et al. 2014. Bayesian modeling of temporal properties of infectious disease in a college student population. *Journal of Applied Statistics* 41, 6 (2014), 1358–1382.
H. Xu et al. 2011. Extracting and integrating data from entire electronic health records for detecting colorectal cancer cases. *AMIA* 2011 (Jan. 2011), 1564–1572.
P. Yadav. 2017. Causal pattern mining in highly heterogeneous and temporal EHRs data. University of Minnesota.
P. Yadav et al. 2015a. Forensic style analysis with survival trajectories. In *Proc. of ICDM*. IEEE, 1069–1074.
P. Yadav et al. 2015b. Modelling trajectories for diabetes complications. In *Proceedings of the 4th Workshop on Data Mining for Medicine and Healthcare. 2015 SIAM International Conference on Data Mining*.
P. Yadav et al. 2016a. Causal inference in observational data. *arXiv preprint arXiv:1611.04660* (2016).
P. Yadav et al. 2016b. Interrogation of bronchoalveolar lavage fluid in acute respiratory distress syndrome using multiplexed proseek immunoassay. In *C49. Respiratory Failure: Clinical and Translational Aspects Of Vili And Lung Protective Mv*. American Thoracic Society, A5246–A5246.
S. Yu et al. 2008. Privacy-preserving Cox regression for survival analysis. In *Proc. of KDD*. ACM, 1034–1042.
H. Zhai et al. 2014. Developing and evaluating a machine learning based algorithm to predict the need of pediatric intensive care unit transfer for newly hospitalized children. *Resuscitation* 85, 8 (Aug. 2014), 1065–1071.
C. Zhang and S. Zhang. 2002. *Association Rule Mining: Models and Algorithms*. Springer-Verlag.
H. Zhang and W. Lu. 2007. Adaptive lasso for Cox's proportional hazards model. *Biometrika* 94, 3 (2007), 691–703.
P. Zhang et al. 2014. Towards personalized medicine: Leveraging patient similarity and drug similarity analytics. *AMIA Summits on Translational Science Proceedings*, vol. 2014. American Medical Informatics Association, 132.
P. Zhang et al. 2015. Label propagation prediction of drug-drug interactions based on clinical side effects. *Scientific Reports* 5 (2015).
W. Zhang et al. 2013. Network-based survival analysis reveals subnetwork signatures for predicting outcomes of ovarian cancer treatment. *PLoS Computational Biology* 9, 3 (2013), e1002975.
D. Zhao et al. 2011. Combining Pubmed knowledge and EHR data to develop a weighted Bayesian network for pancreatic cancer prediction. *JBI* 44, 5 (2011), 859–868.
K. Zolfaghar et al. 2013. Risk-o-meter: An intelligent clinical risk calculator. In *Proc. KDD*. ACM, 1518–1521.

Received March 2017; revised July 2017; accepted July 2017